

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
    nan_ratio = df[cols_to_check].isna().mean()
    valid_features = nan_ratio[nan_ratio <= 0.15].index
    valid_features_list = list(valid_features)
```

```

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()

```

```

        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r"):
            #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
                dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}

```

```

elif camp == "2026_Gen":
    #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
    #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
    mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
else:
    mapping = {}

return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test

- `train_cv`: mean score train of the CV
- `test_cv`: mean score test of the CV
- `test`: score on the testing dataset
- `diff`: `test-test_cv`
- `r_test`: `test/test_cv`
- `r_train`: `train_cv/test_cv`

Colors:

[Orange] `r_test` < 0.95 (under performance of the test)

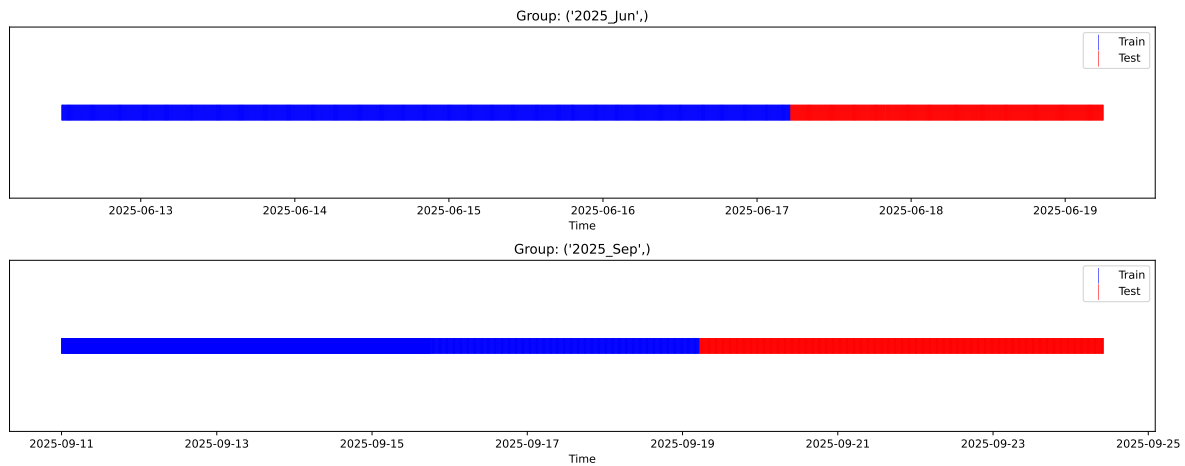
[Green] `r_test` > 1.05 (over performance of the test)

[Red] `r_train` not in [0.95, 1.05] (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: `X_train` vs `X_test`

External Split: Train: 5719 samples Test: 2449 samples



2.2.2 Internal CV Folds

Fold 1/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



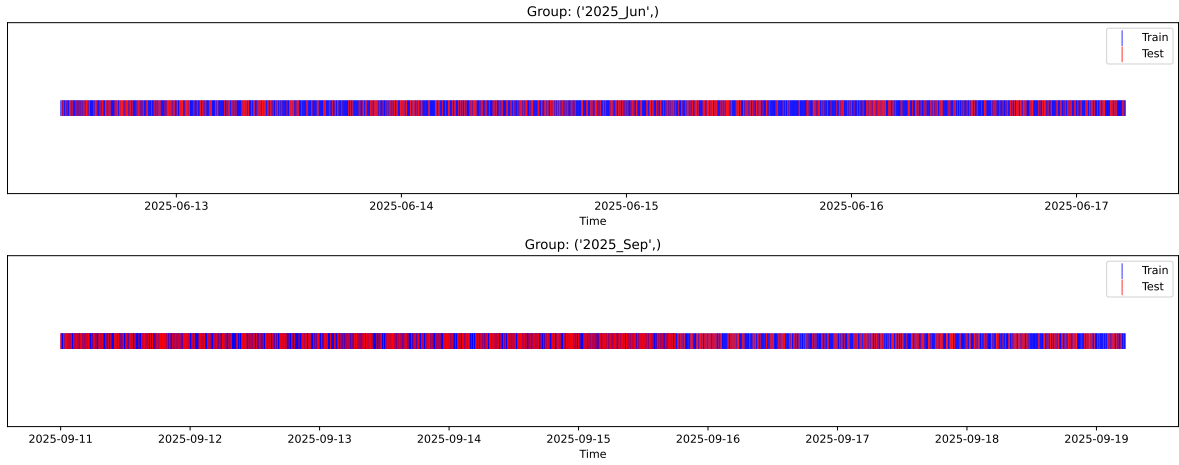
Fold 2/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



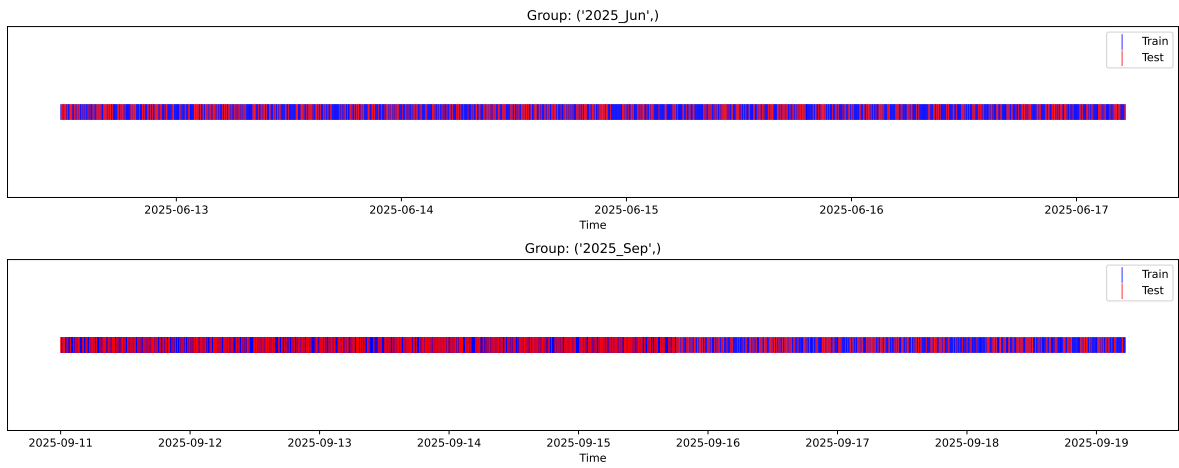
Fold 3/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



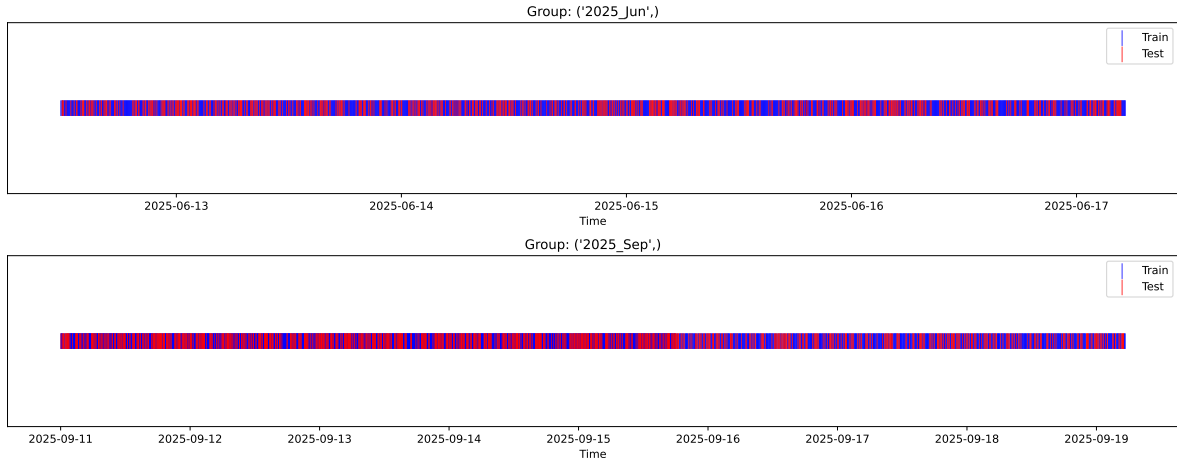
Fold 4/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.354, std=0.804, min=0.959, max=8.682
 y_{test} : mean=2.340, std=0.772, min=1.071, max=13.845

Fold 0 — y_{test} : mean=2.35, std=0.79, y_{train} : mean=2.35, std=0.81

Fold 1 — y_{test} : mean=2.37, std=0.79, y_{train} : mean=2.35, std=0.81

Fold 2 — y_{test} : mean=2.36, std=0.80, y_{train} : mean=2.35, std=0.81

Fold 3 — y_{test} : mean=2.33, std=0.82, y_{train} : mean=2.36, std=0.80

Fold 4 — y_{test} : mean=2.33, std=0.77, y_{train} : mean=2.36, std=0.82

train scores CV: [0.40125516 0.40930372 0.40130967 0.40621486 0.39718676]

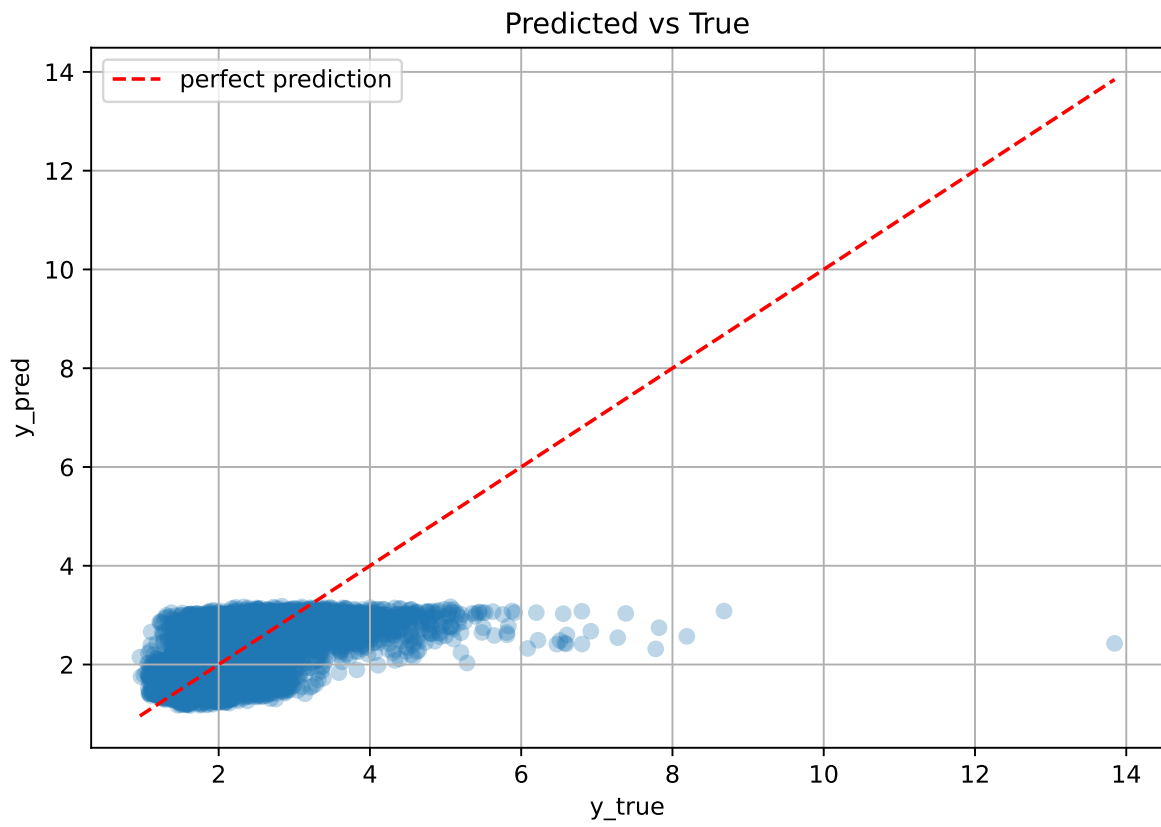
test scores CV: [0.41310399 0.39293914 0.41249368 0.39999369 0.42378683]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4031	0.4085	0.0108	0.1872	0.1866	-0.2219	0.4568	0.9868

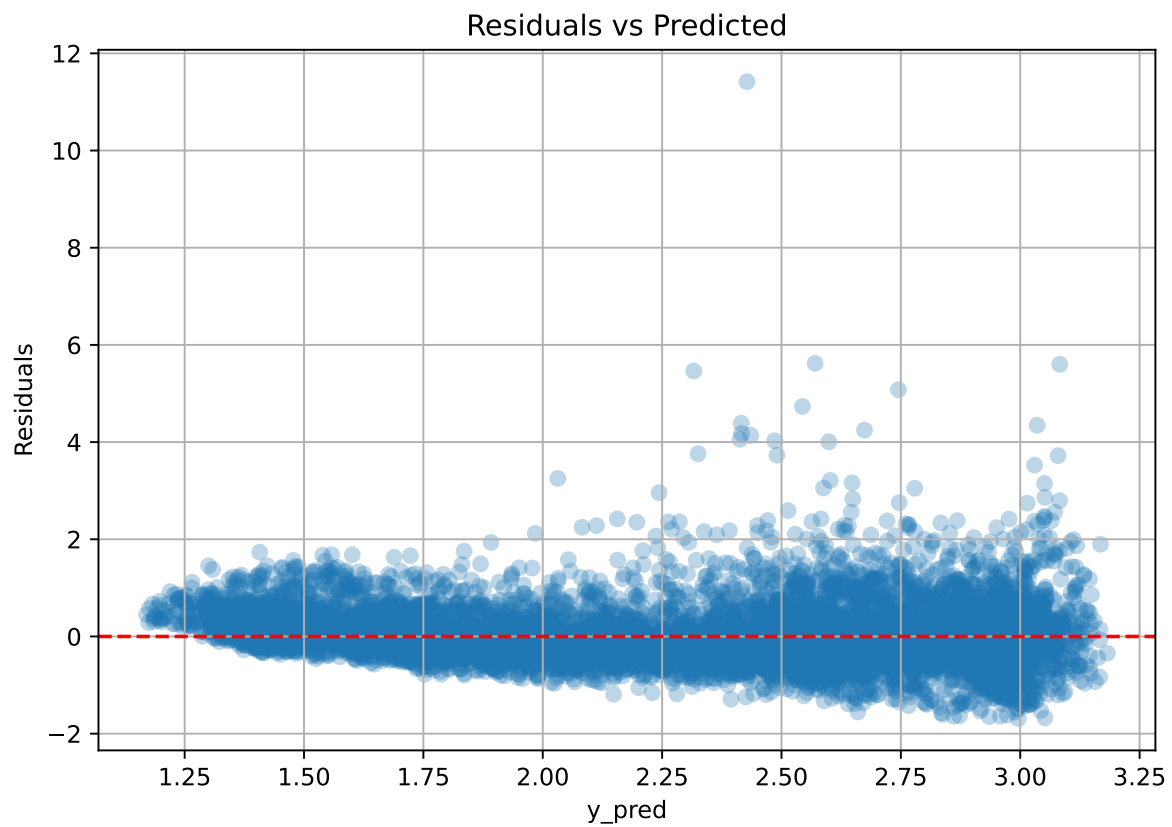
Coefficients:

coef_r_CO2 = -0.0195, coef_r_NH3 = -0.0421, coef_r_THI = 0.8947, coef_r_temperature = -1.1610, coef_r_umidity = -0.8420, intercept = 2.3535,

2.3.2 Scatter



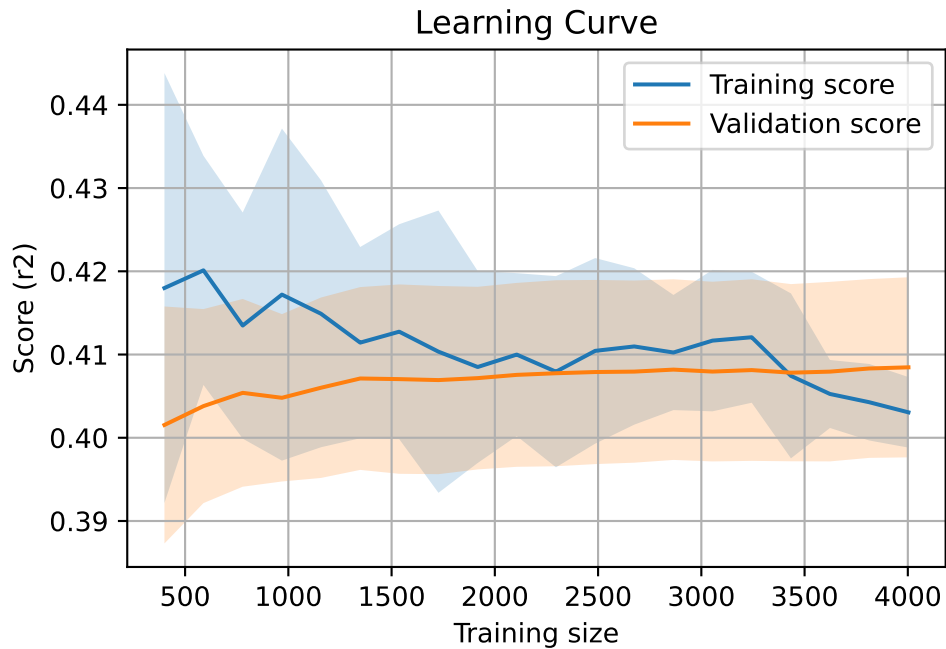
2.3.3 Scatter residuals



2.3.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450

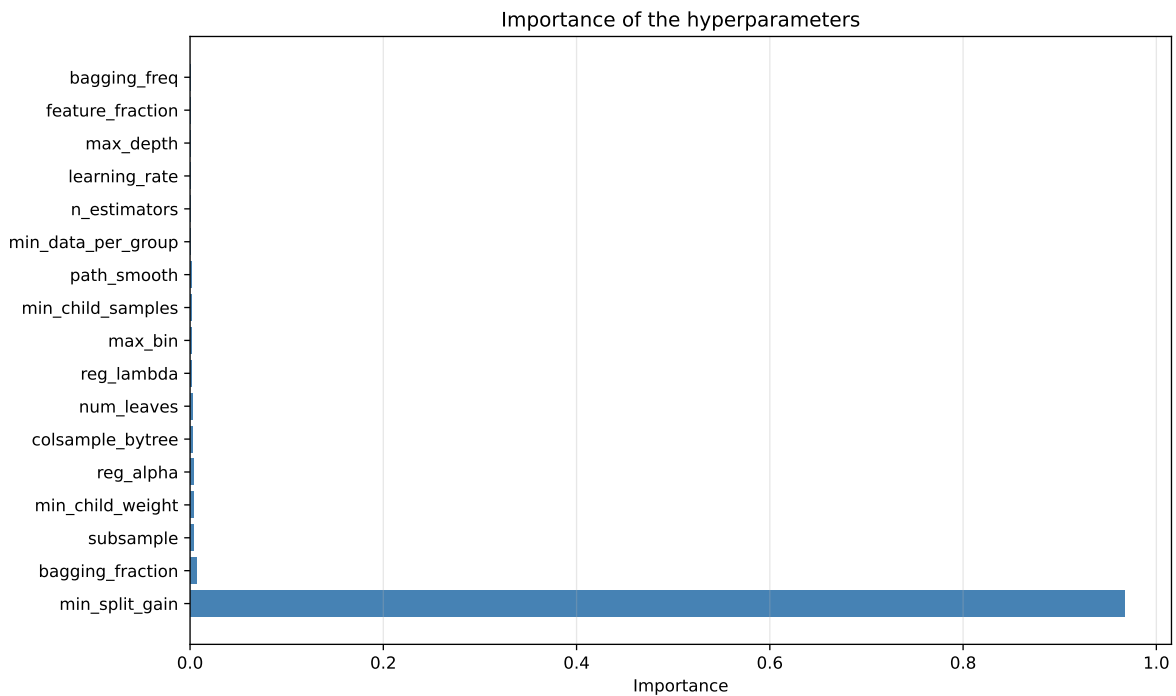
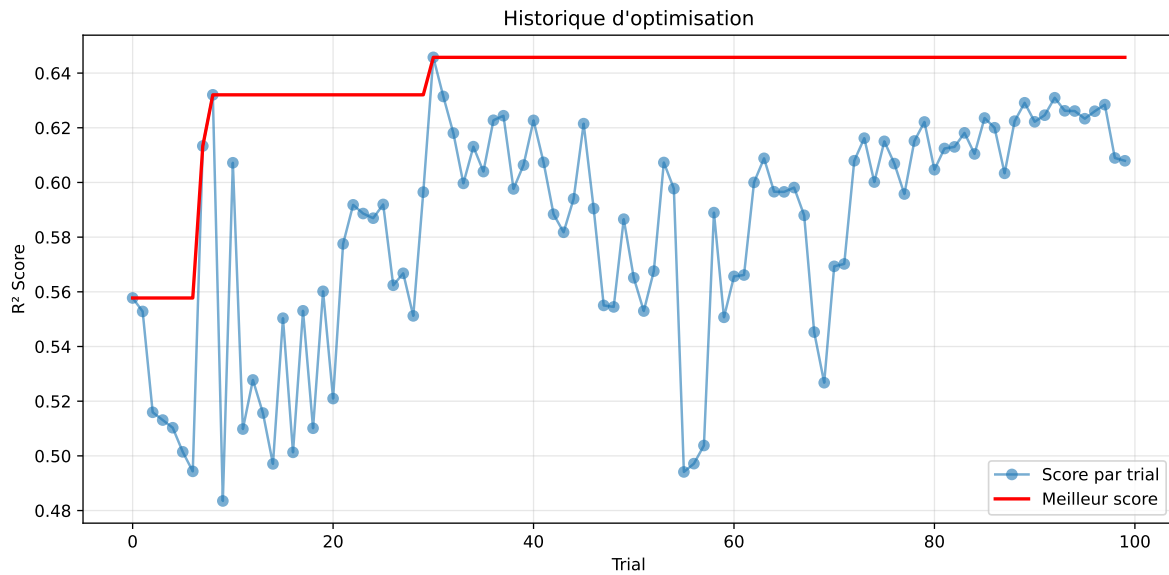


2.4 Model : LGBM

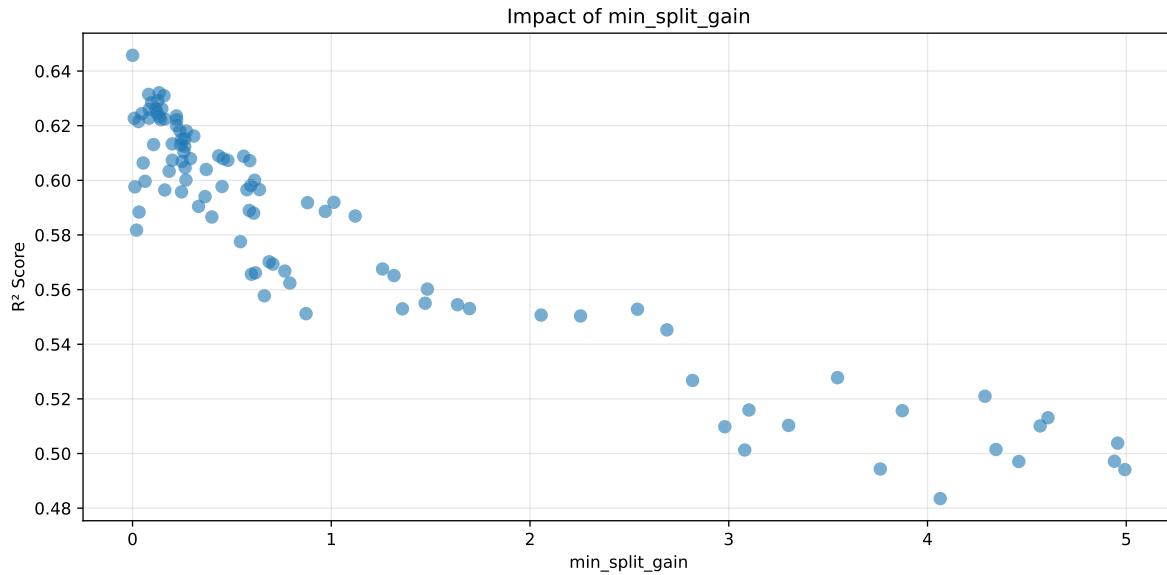
2.4.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.354, std=0.804, min=0.959, max=8.682
y_test: mean=2.340, std=0.772, min=1.071, max=13.845

===== TOP Importance FEATURES =====
feature importance r_CO2 2126 r_umidity 2072 r_NH3 1957 r_THI 1909 r_temperature
1757



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7774	0.6458	0.1795	0.014	0.6458	0.1516	-0.4942	0.2347	1.2039
2	0.7757	0.632	0.2027	0.0185	0.632	0.1847	-0.4474	0.2922	1.2273
3	0.7621	0.6315	0.2093	0.0175	0.6315	0.2089	-0.4226	0.3307	1.2068
4	0.7515	0.631	0.2104	0.0181	0.631	0.1856	-0.4453	0.2942	1.191
5	0.7246	0.6291	0.2238	0.02	0.6291	0.2214	-0.4077	0.352	1.1517
6	0.7206	0.6285	0.2259	0.0212	0.6285	0.2152	-0.4132	0.3424	1.1467
7	0.7373	0.6262	0.2152	0.0184	0.6262	0.2119	-0.4143	0.3383	1.1775
8	0.7272	0.6261	0.2117	0.0183	0.6261	0.1761	-0.4501	0.2812	1.1614
9	0.7251	0.6261	0.2288	0.023	0.6261	0.2261	-0.3999	0.3612	1.1581
10	0.7234	0.6246	0.2212	0.0171	0.6246	0.2134	-0.4112	0.3417	1.1582
---	---	---	---	---	---	---	---	---	---
100	0.4984	0.4835	0.2726	0.0105	0.4835	0.2696	-0.214	0.5575	1.0308

Hyperparameters:

Rank 1: {'reg__n_estimators': 300, 'reg__max_depth': 7, 'reg__learning_rate': 0.09277311195226824, 'reg__num_leaves': 62, 'reg__subsample': 0.7055049629491035, 'reg__colsample_bytree': 0.8965026437107444, 'reg__min_child_samples': 33, 'reg__reg_alpha': 0.7249684304949975, 'reg__reg_lambda': 46.031313265128446, 'reg__min_split_gain': 0.0004675098260185917, 'reg__path_smooth': 0.1681503311821948, 'reg__min_child_weight': 0.009742662419382472, 'reg__feature_fraction': 0.5317370490068216, 'reg__bagging_fraction': 0.6380875262342125, 'reg__bagging_freq': 6, 'reg__max_bin': 91, 'reg__min_data_per_group': 53}

Rank 2: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.070827626649512, 'reg__num_leaves': 20, 'reg__subsample': 0.5498837289589729, 'reg__colsample_bytree': 0.7998398353775739, 'reg__min_child_samples': 12, 'reg__reg_alpha': 0.9567132368060649, 'reg__reg_lambda': 30.930824977829012, 'reg__min_split_gain': 0.13393318743822713, 'reg__path_smooth': 4.396082109341663, 'reg__min_child_weight': 0.04323841096483267, 'reg__feature_fraction': 0.8297407104402794, 'reg__bagging_fraction': 0.5061379750372269, 'reg__bagging_freq': 4, 'reg__max_bin': 163, 'reg__min_data_per_group': 153}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.09249397344924723, 'reg__num_leaves': 19, 'reg__subsample': 0.71184953186867, 'reg__colsample_bytree': 0.8886067242065413, 'reg__min_child_samples': 34, 'reg__reg_alpha': 0.6515404892917429, 'reg__reg_lambda': 45.950828679237375, 'reg__min_split_gain': 0.08097391890588163, 'reg__path_smooth': 1.9762688561315658, 'reg__min_child_weight': 0.009408560784266554, 'reg__feature_fraction': 0.5167775112244374, 'reg__bagging_fraction': 0.8227814654861663, 'reg__bagging_freq': 6, 'reg__max_bin': 206, 'reg__min_data_per_group': 121}

Rank 4: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.08015951104055773, 'reg__num_leaves': 22, 'reg__subsample': 0.5032594805749723, 'reg__colsample_bytree': 0.674054895802772, 'reg__min_child_samples': 10, 'reg__reg_alpha': 1.2252065275103377, 'reg__reg_lambda': 12.795556321387807, 'reg__min_split_gain': 0.1582270581327263, 'reg__path_smooth': 3.2233467897540558, 'reg__min_child_weight': 0.0011513383954322093, 'reg__feature_fraction': 0.6303039679565486, 'reg__bagging_fraction': 0.5398999099266379, 'reg__bagging_freq': 6, 'reg__max_bin': 99, 'reg__min_data_per_group': 139}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.08472383063620934, 'reg__num_leaves': 22, 'reg__subsample': 0.7519748028730142, 'reg__colsample_bytree': 0.6794428337901062, 'reg__min_child_samples': 19, 'reg__reg_alpha': 1.2969719008230818, 'reg__reg_lambda': 23.955115267374207, 'reg__min_split_gain': 0.12742133144935885, 'reg__path_smooth': 2.8819584221750976, 'reg__min_child_weight': 0.001119561829790015, 'reg__feature_fraction': 0.5881878149072582, 'reg__bagging_fraction': 0.8923862894898769, 'reg__bagging_freq': 6, 'reg__max_bin': 107, 'reg__min_data_per_group': 148}

Rank 6: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.08821308749712668, 'reg__num_leaves': 24, 'reg__subsample': 0.5021522043011537, 'reg__colsample_bytree': 0.5001697645660726, 'reg__min_child_samples': 17, 'reg__reg_alpha': 1.821136747311924, 'reg__reg_lambda': 29.581520044856063, 'reg__min_split_gain': 0.09623201604147026, 'reg__path_smooth': 2.8179492986299817, 'reg__min_child_weight': 0.004560344436181619, 'reg__feature_fraction': 0.5580583946667461, 'reg__bagging_fraction': 0.8999108363732597, 'reg__bagging_freq': 5, 'reg__max_bin': 110, 'reg__min_data_per_group': 156}

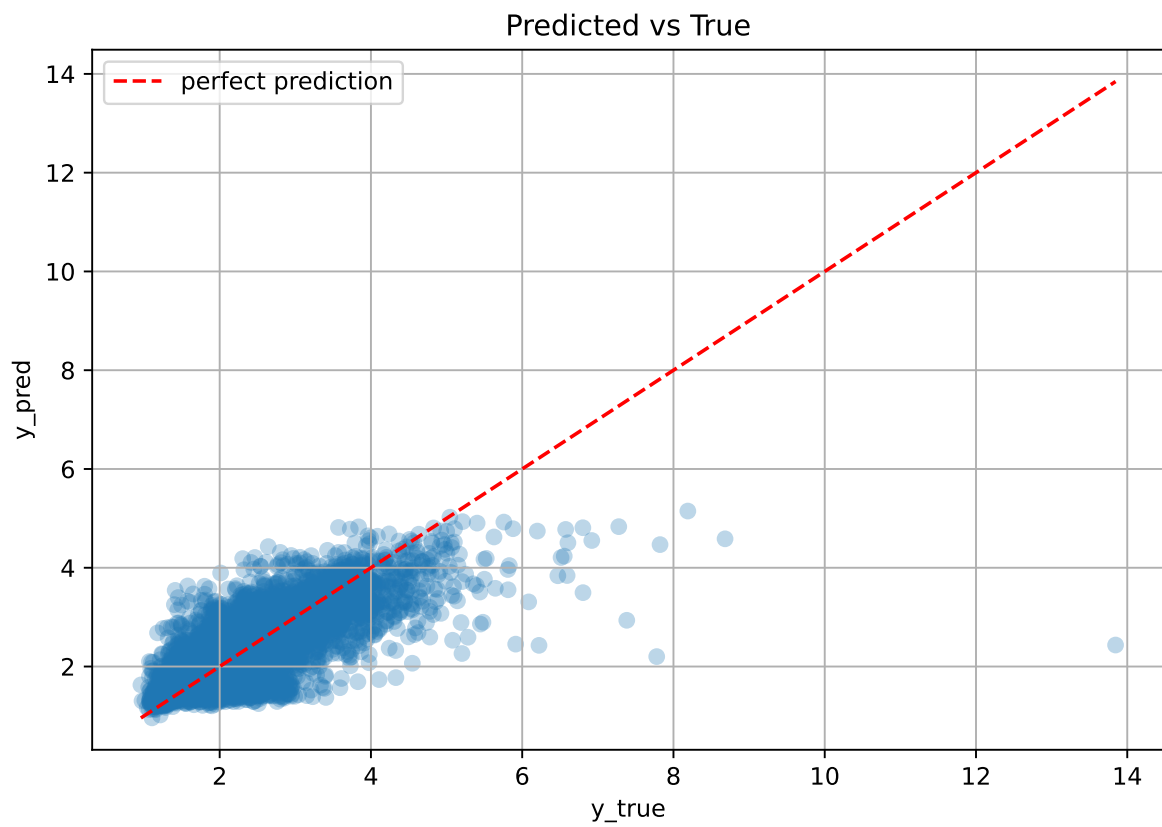
Rank 7: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.08385405410604986, 'reg__num_leaves': 22, 'reg__subsample': 0.5040494729321563, 'reg__colsample_bytree': 0.5125445701677034, 'reg__min_child_samples': 11, 'reg__reg_alpha': 1.273982530861017, 'reg__reg_lambda': 22.858355735838884, 'reg__min_split_gain': 0.14727603482086174, 'reg__path_smooth': 3.165326269940076, 'reg__min_child_weight': 0.0010311648944364787, 'reg__feature_fraction': 0.6227669437768114, 'reg__bagging_fraction': 0.5463624968037715, 'reg__bagging_freq': 6, 'reg__max_bin': 111, 'reg__min_data_per_group': 154}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.08278618681006354, 'reg__num_leaves': 22, 'reg__subsample': 0.6701938770751276, 'reg__colsample_bytree': 0.6848610829572666, 'reg__min_child_samples': 18, 'reg__reg_alpha': 2.1554367360739928, 'reg__reg_lambda': 9.395460824583829, 'reg__min_split_gain': 0.11793093779823592, 'reg__path_smooth': 3.2049449287359266, 'reg__min_child_weight': 0.02050022134191707, 'reg__feature_fraction': 0.5594317062086869, 'reg__bagging_fraction': 0.5419818568177905, 'reg__bagging_freq': 6, 'reg__max_bin': 99, 'reg__min_data_per_group': 150}

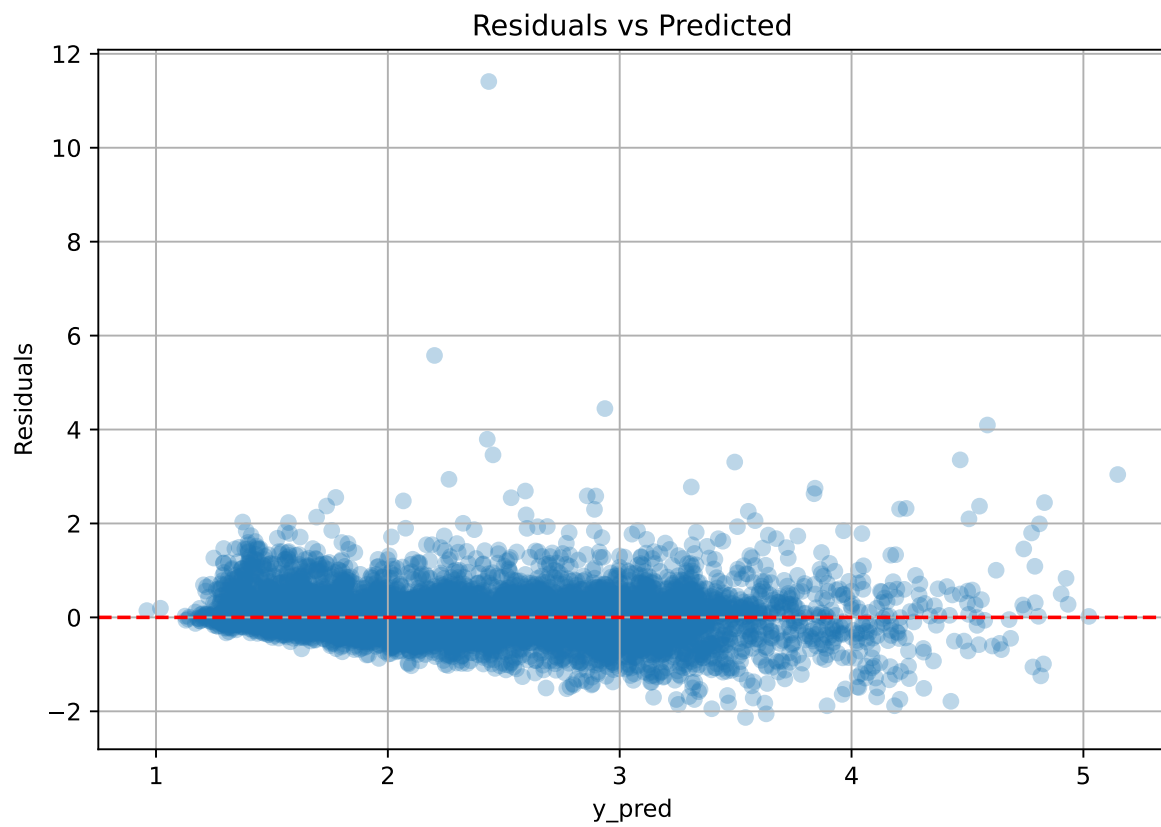
Rank 9: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.08702179658636891, 'reg__num_leaves': 16, 'reg__subsample': 0.5526232203039089, 'reg__colsample_bytree': 0.5259892157309536, 'reg__min_child_samples': 19, 'reg__reg_alpha': 2.168389118135611, 'reg__reg_lambda': 24.07334685545162, 'reg__min_split_gain': 0.08450481296322002, 'reg__path_smooth': 2.636993334671121, 'reg__min_child_weight': 0.004317751105570671, 'reg__feature_fraction': 0.5573824835081133, 'reg__bagging_fraction': 0.8396934050363563, 'reg__bagging_freq': 5, 'reg__max_bin': 112, 'reg__min_data_per_group': 138}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.04741358956147268, 'reg__num_leaves': 25, 'reg__subsample': 0.6710214193759165, 'reg__colsample_bytree': 0.598846636676369, 'reg__min_child_samples': 10, 'reg__reg_alpha': 2.1686499182577856, 'reg__reg_lambda': 38.91534224068234, 'reg__min_split_gain': 0.12380441947782286, 'reg__path_smooth': 3.1976727136957495, 'reg__min_child_weight': 0.022850568171407307, 'reg__feature_fraction': 0.5579001543494309, 'reg__bagging_fraction': 0.5535189336975876, 'reg__bagging_freq': 6, 'reg__max_bin': 106, 'reg__min_data_per_group': 138}

2.4.2 Scatter



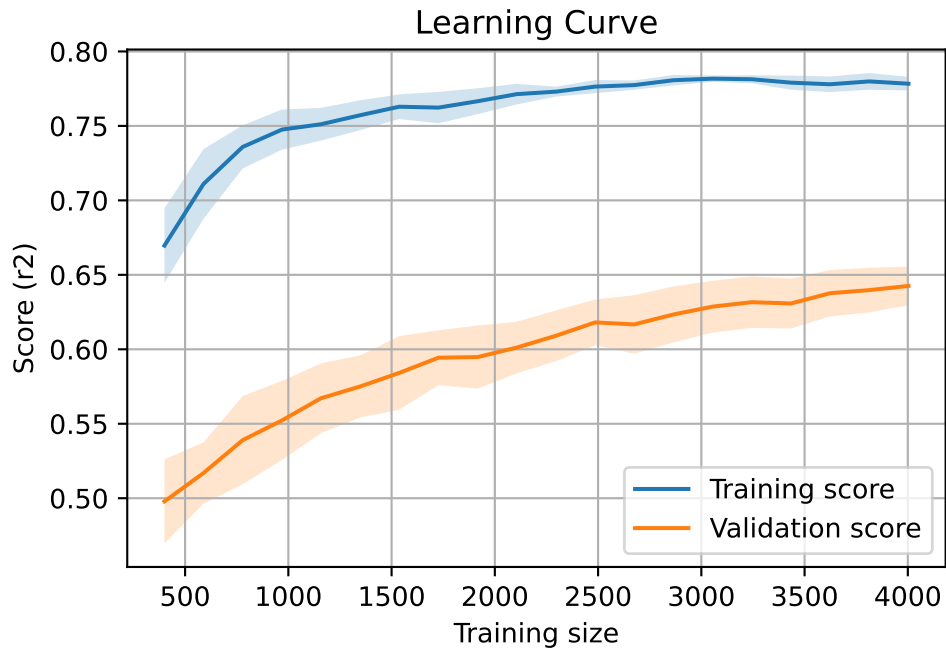
2.4.3 Scatter residuals



2.4.4 Learning curve

`Len(group)` : 8168, `Len(training)` : 5718,

`Len(training_real)` : 4003, `Len(test_of_training)` : 1715, `Len(test)` : 2450

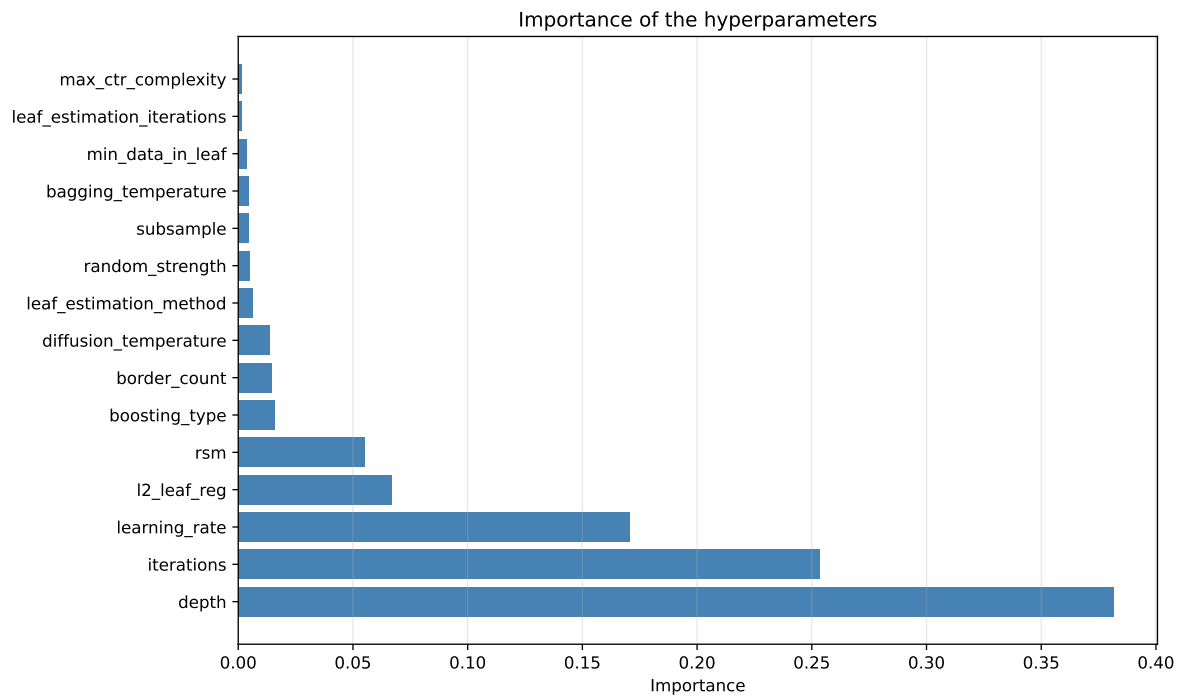
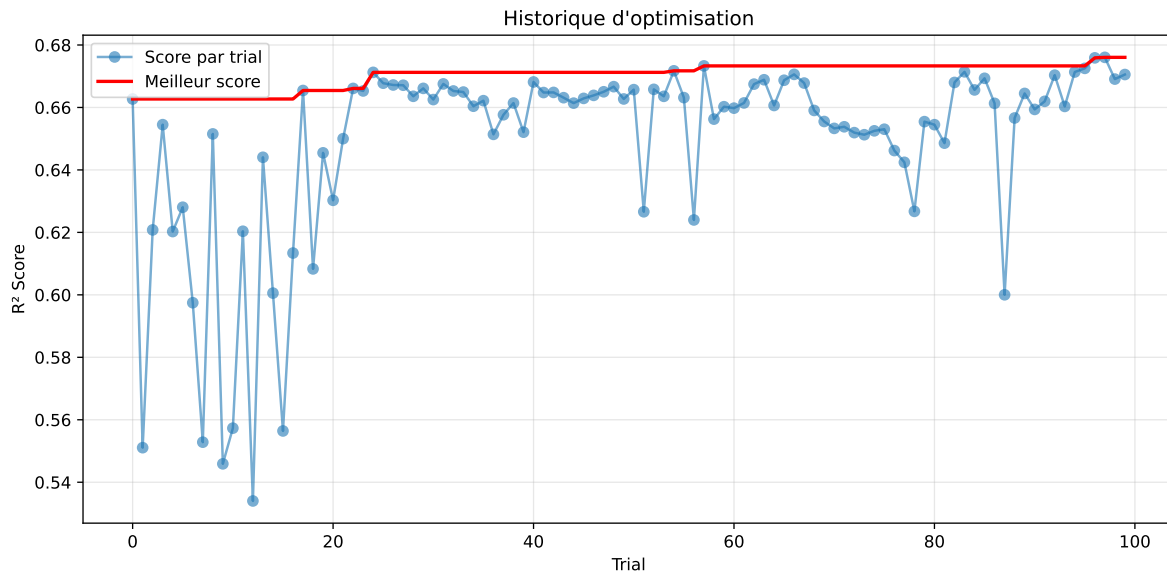


2.5 Model : CatBoost

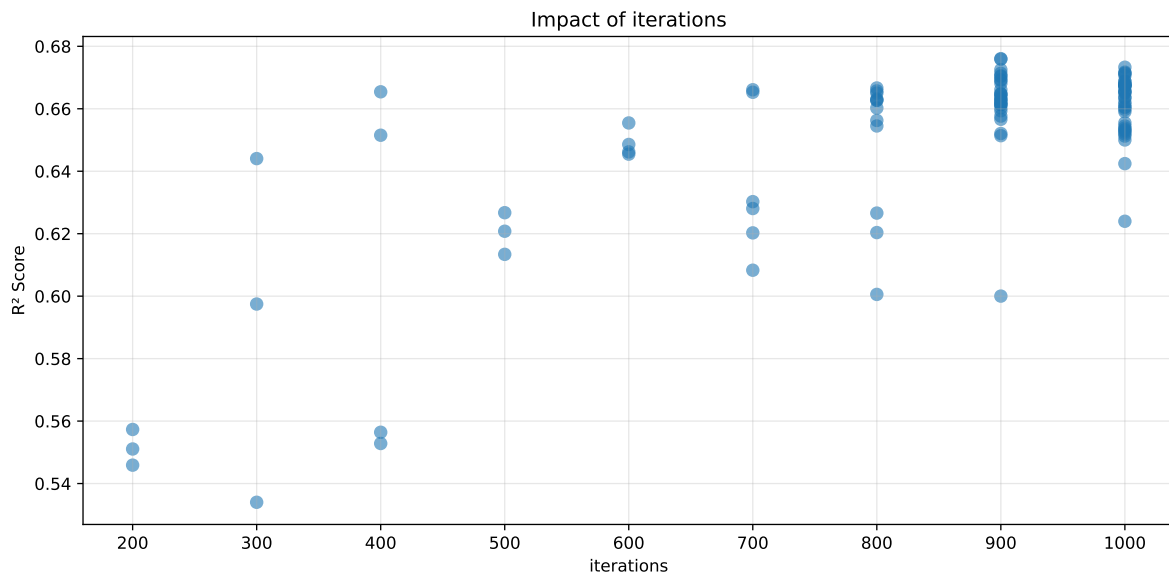
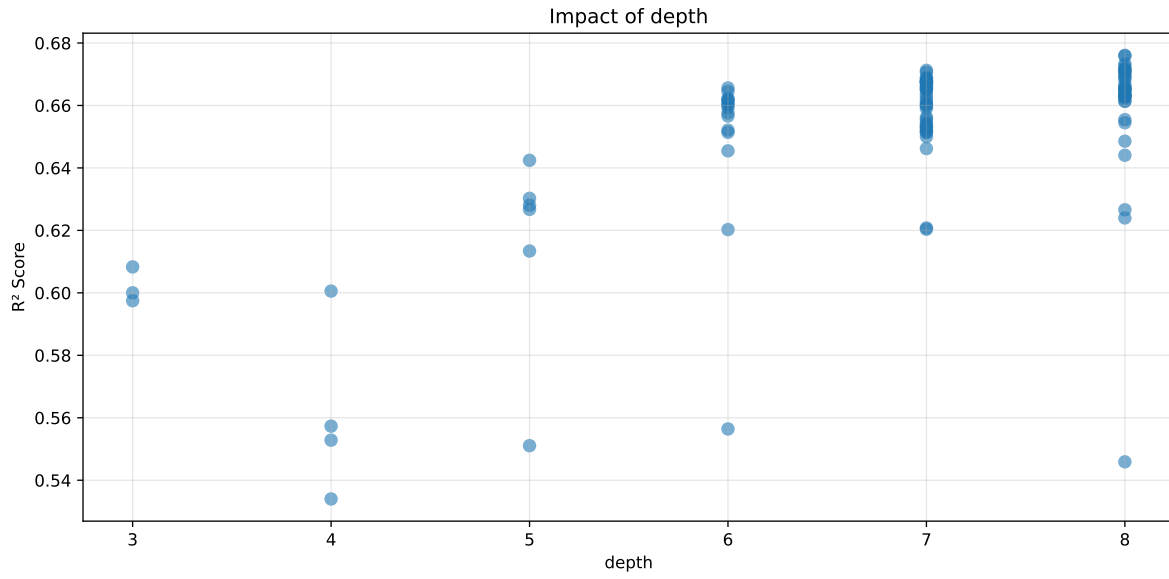
2.5.1 Gridsearch

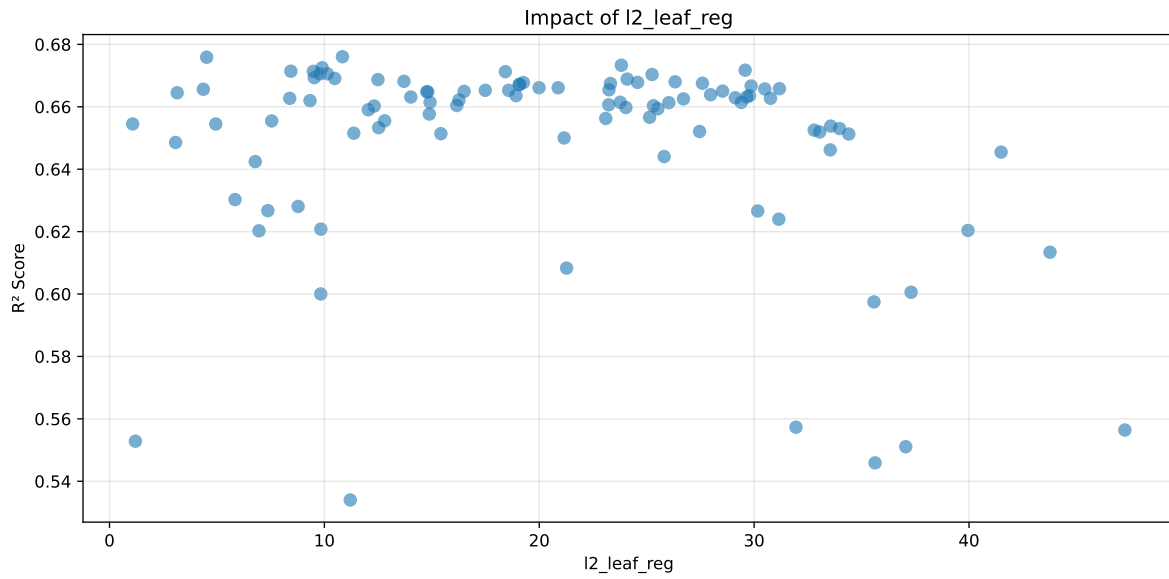
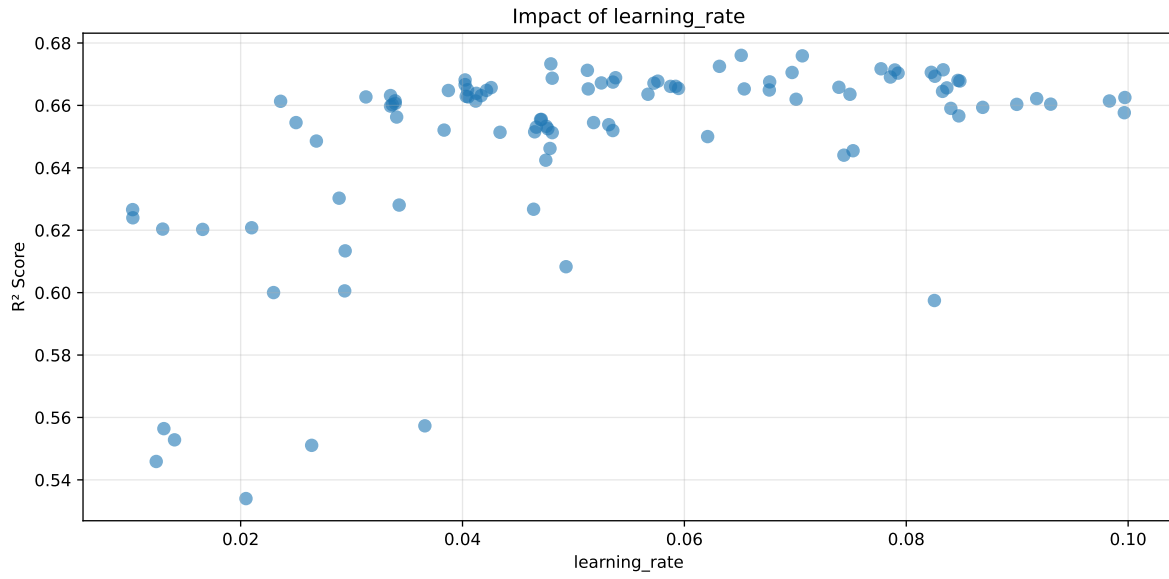
Distribution y_train vs y_test: y_train: mean=2.354, std=0.804, min=0.959, max=8.682
y_test: mean=2.340, std=0.772, min=1.071, max=13.845

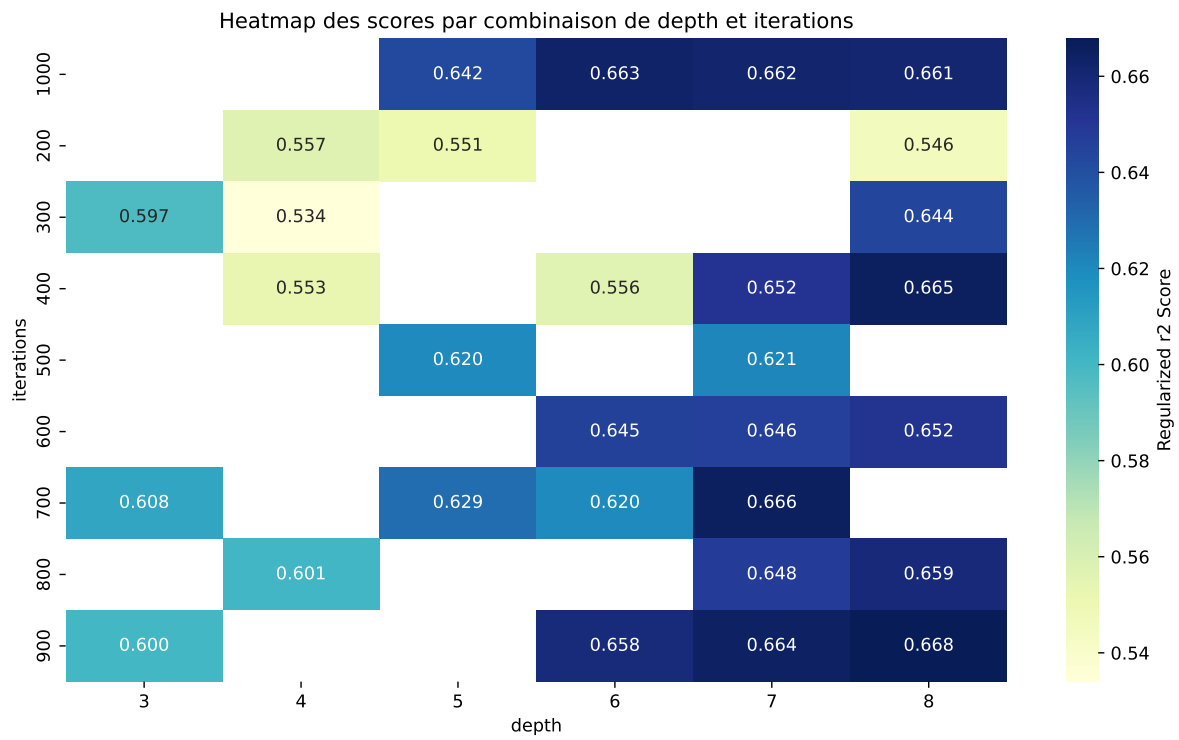
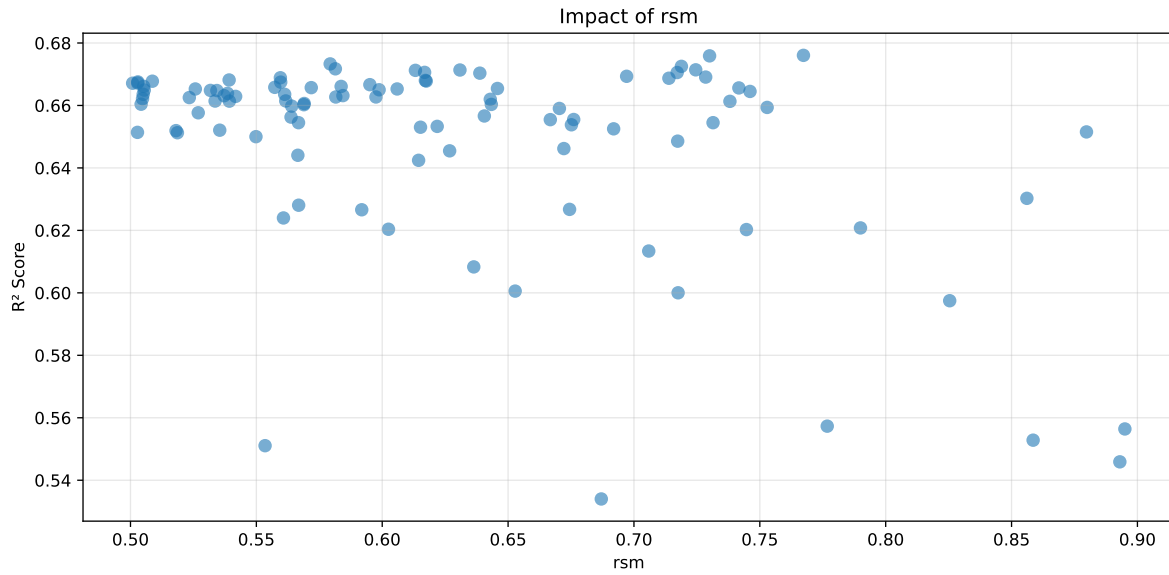
===== TOP Importance FEATURES =====
feature importance r_umidity 26.951742 r_NH3 24.488840 r_CO2 22.286110 r_THI 15.469828
r_temperature 10.803481



Paramètres avec >5% d'importance : ['depth', 'iterations', 'learning_rate', 'l2_leaf_reg', 'rsm']







rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.9348	0.676	0.1603	0.0221	0.676	0.1598	-0.5162	0.2364	1.3828
2	0.9514	0.6759	0.1602	0.0218	0.6759	0.1579	-0.5179	0.2337	1.4077
3	0.9011	0.6733	0.1818	0.0197	0.6733	0.185	-0.4883	0.2748	1.3383
4	0.9335	0.6725	0.1656	0.02	0.6725	0.1711	-0.5014	0.2544	1.3881
5	0.9248	0.6717	0.1427	0.0205	0.6717	0.1289	-0.5428	0.1919	1.3767
6	0.9567	0.6714	0.1418	0.0191	0.6714	0.1346	-0.5368	0.2005	1.4249
7	0.9435	0.6714	0.1512	0.0185	0.6714	0.1367	-0.5347	0.2036	1.4054
8	0.8743	0.6712	0.1816	0.0169	0.6712	0.1784	-0.4929	0.2658	1.3025
9	0.9186	0.6706	0.1478	0.0208	0.6706	0.138	-0.5326	0.2058	1.3698
10	0.9396	0.6705	0.1503	0.022	0.6705	0.1457	-0.5249	0.2173	1.4012
---	---	---	---	---	---	---	---	---	---
100	0.5517	0.534	0.2898	0.0189	0.534	0.2902	-0.2438	0.5434	1.0331

Hyperparameters:

Rank 1: {'reg__iterations': 900, 'reg__depth': 8, 'reg__learning_rate': 0.06512226927560569, 'reg__l2_leaf_reg': 10.840898825686729, 'reg__bagging_temperature': 2.51571259554781, 'reg__random_strength': 2.238306250397736, 'reg__subsample': 0.6435241260569061, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7673314047491697, 'reg__border_count': 83, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 1052.84960730107}

Rank 2: {'reg__iterations': 900, 'reg__depth': 8, 'reg__learning_rate': 0.070631478867872, 'reg__l2_leaf_reg': 4.520778893124319, 'reg__bagging_temperature': 2.5081828504117185, 'reg__random_strength': 2.1686678724644093, 'reg__subsample': 0.7220075255945881, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.73004191678665, 'reg__border_count': 95, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 6176.720273021694}

Rank 3: {'reg__iterations': 1000, 'reg__depth': 8, 'reg__learning_rate': 0.04796209116028225, 'reg__l2_leaf_reg': 23.821224016725825, 'reg__bagging_temperature': 2.122626885229689, 'reg__random_strength': 3.9913102525672275, 'reg__subsample': 0.687548378251743, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.5793259327779916, 'reg__border_count': 59, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 5910.290232699099}

Rank 4: {'reg__iterations': 900, 'reg__depth': 8, 'reg__learning_rate': 0.06316037396007092, 'reg__l2_leaf_reg': 9.897932559119386, 'reg__bagging_temperature': 2.0903921911789927, 'reg__random_strength': 2.1739022336428193, 'reg__subsample': 0.6418548301869791, 'reg__min_data_in_leaf': 32, 'reg__leaf_estimation_iterations': 6, 'reg__rsm':

0.7188508575086977, 'reg__border_count': 89, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4194.880535571732}

Rank 5: {'reg__iterations': 1000, 'reg__depth': 8, 'reg__learning_rate': 0.07772682288975052, 'reg__l2_leaf_reg': 29.583457735269132, 'reg__bagging_temperature': 3.612025443206293, 'reg__random_strength': 1.1906454584115493, 'reg__subsample': 0.5636380463446111, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.5813859126651781, 'reg__border_count': 38, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7270.16101912231}

Rank 6: {'reg__iterations': 1000, 'reg__depth': 8, 'reg__learning_rate': 0.08332480443374315, 'reg__l2_leaf_reg': 8.440898264286325, 'reg__bagging_temperature': 2.7811201193082336, 'reg__random_strength': 1.907987580005995, 'reg__subsample': 0.5390428803039856, 'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7245401134634634, 'reg__border_count': 88, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 1562.8596548576352}

Rank 7: {'reg__iterations': 900, 'reg__depth': 8, 'reg__learning_rate': 0.07896883673636468, 'reg__l2_leaf_reg': 9.488714636630247, 'reg__bagging_temperature': 2.0946206103127096, 'reg__random_strength': 1.6659636232305886, 'reg__subsample': 0.6543788582873273, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6308776420426261, 'reg__border_count': 53, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 7150.34125568242}

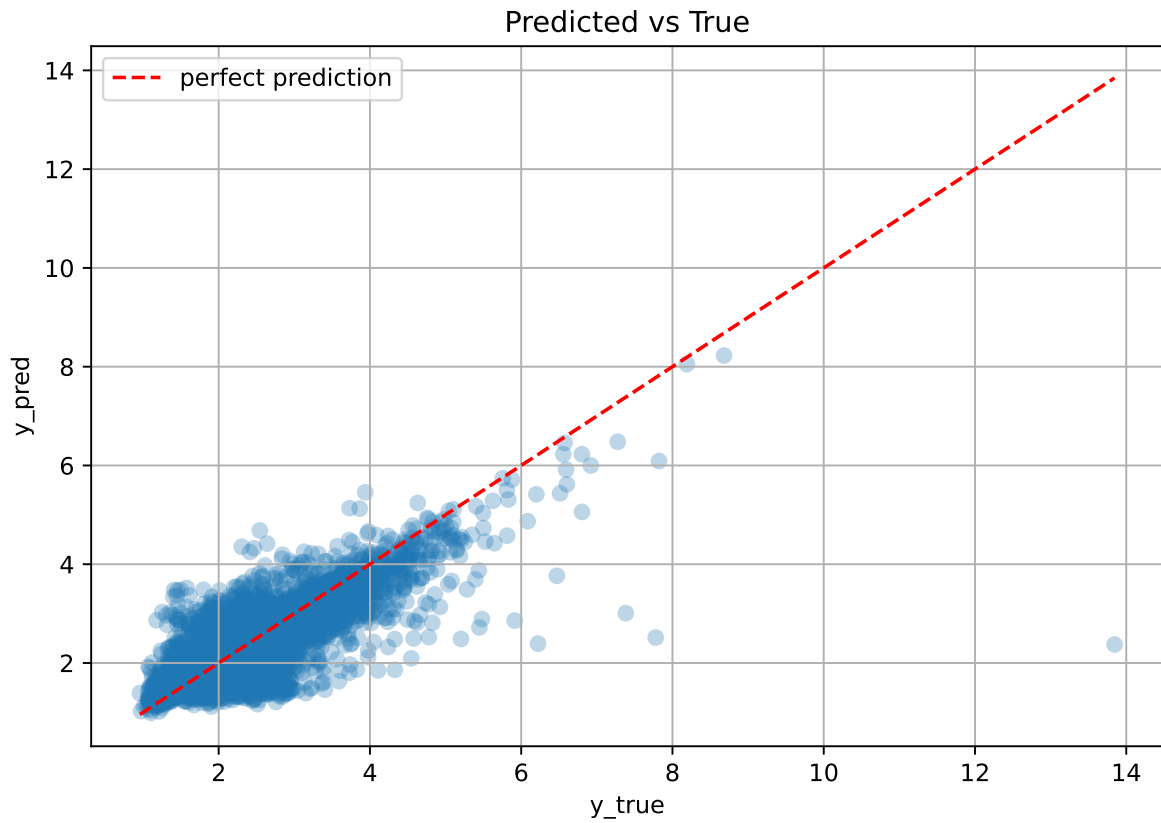
Rank 8: {'reg__iterations': 1000, 'reg__depth': 7, 'reg__learning_rate': 0.05125566656507743, 'reg__l2_leaf_reg': 18.423182828310637, 'reg__bagging_temperature': 1.1963691608710847, 'reg__random_strength': 1.1031386576261295, 'reg__subsample': 0.5106098570018802, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.6131854026774829, 'reg__border_count': 33, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 7418.22066611627}

Rank 9: {'reg__iterations': 1000, 'reg__depth': 7, 'reg__learning_rate': 0.08225508623832684, 'reg__l2_leaf_reg': 10.133957712986334, 'reg__bagging_temperature': 3.545567257591944, 'reg__random_strength': 1.8011264562098854, 'reg__subsample': 0.5626144462045509, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.616838803882283, 'reg__border_count': 36, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 6236.169945820853}

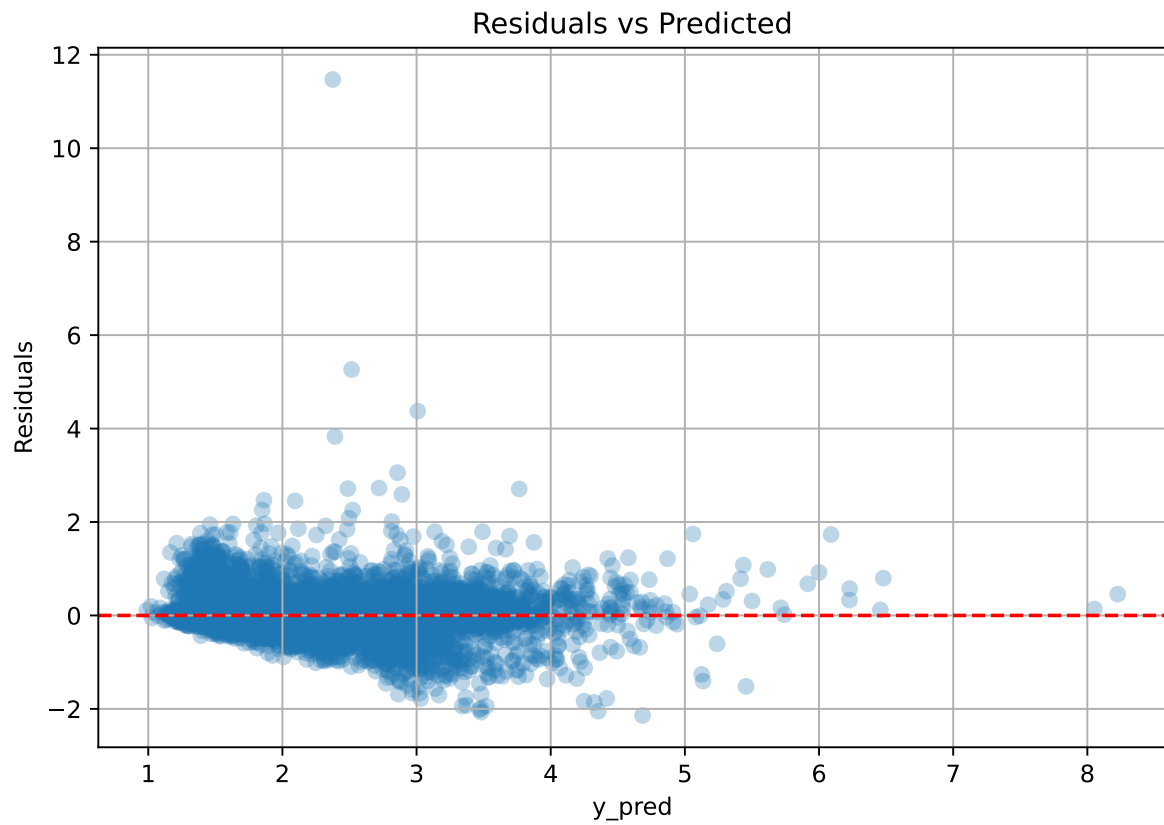
Rank 10: {'reg__iterations': 900, 'reg__depth': 8, 'reg__learning_rate': 0.06970758359083637, 'reg__l2_leaf_reg': 9.790941269481806, 'reg__bagging_temperature': 2.487177822416034,

'reg__random_strength': 0.709895552941469, 'reg__subsample': 0.6408961246227645,
'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 6, 'reg__rsm':
0.717171282398252, 'reg__border_count': 55, 'reg__leaf_estimation_method': 'Gradient',
'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature':
5429.101143843766}

2.5.2 Scatter



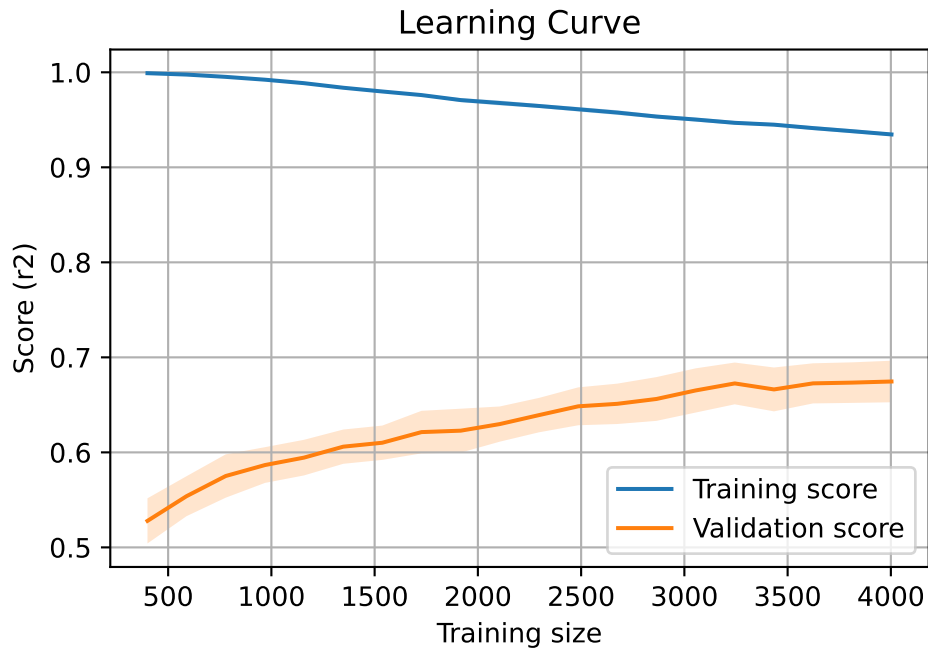
2.5.3 Scatter residuals



2.5.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450



2.6 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.403	0.408	0.011	0.187	0.187	-0.222	0.457	0.987	nan
LGBM	0.777	0.646	0.014	0.18	0.152	-0.494	0.235	1.204	0.6458
CatBoost	0.935	0.676	0.022	0.16	0.16	-0.516	0.236	1.383	0.676

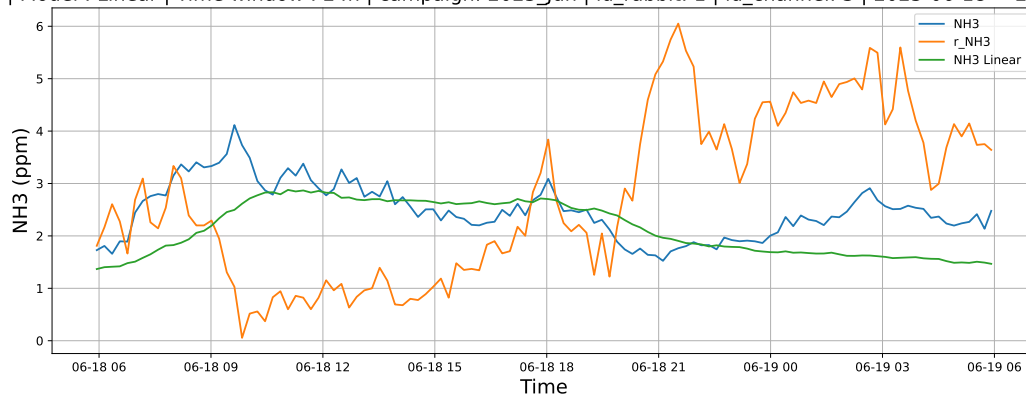
3 Plot

3.1 Standalone plots

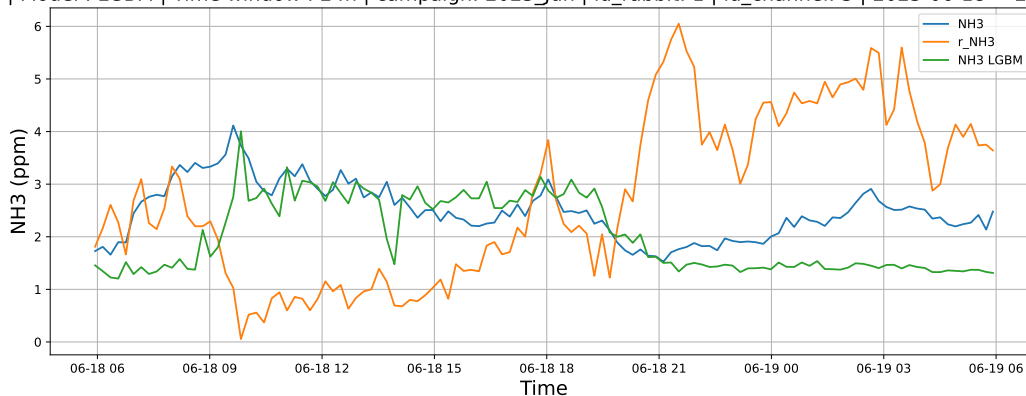
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

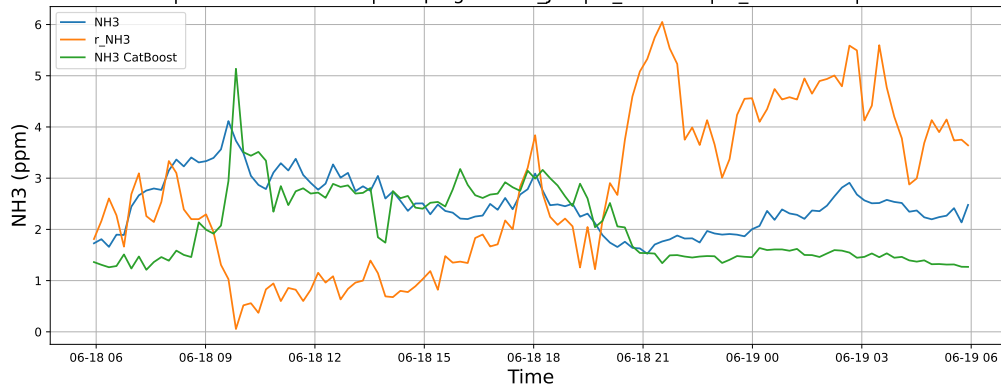
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

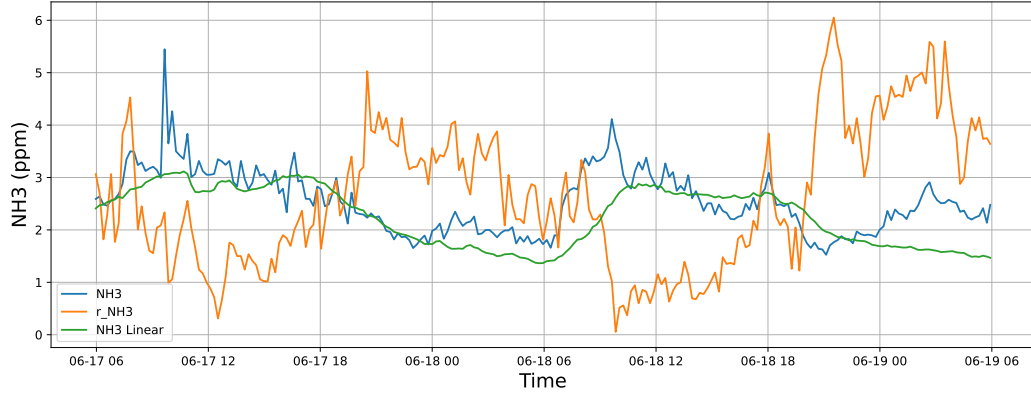


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

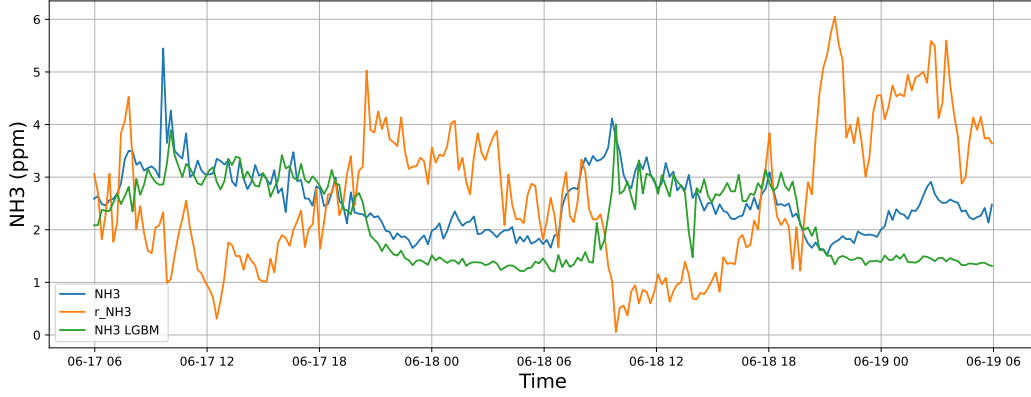


3.1.1.2 Time window : 48

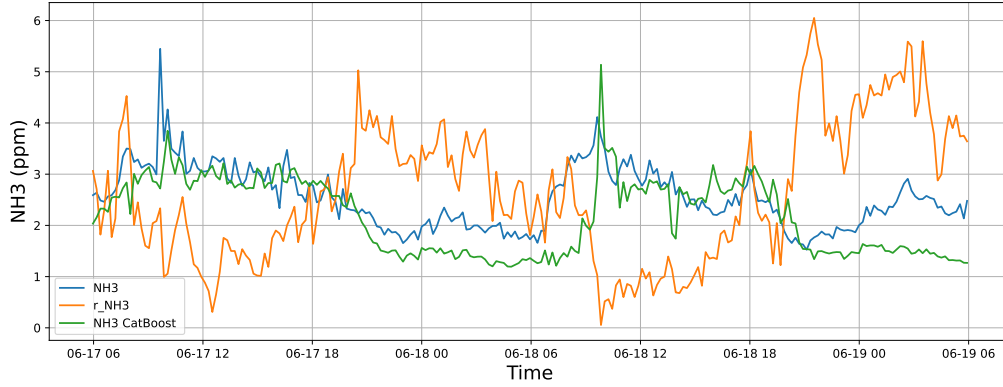
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

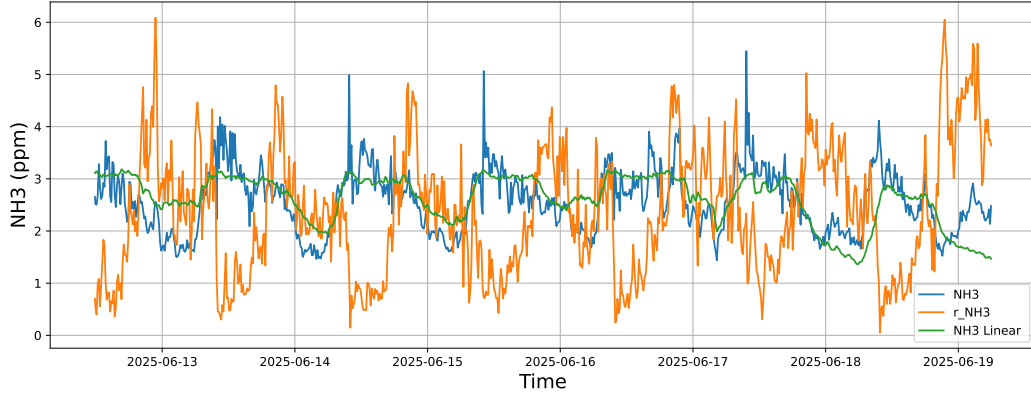


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

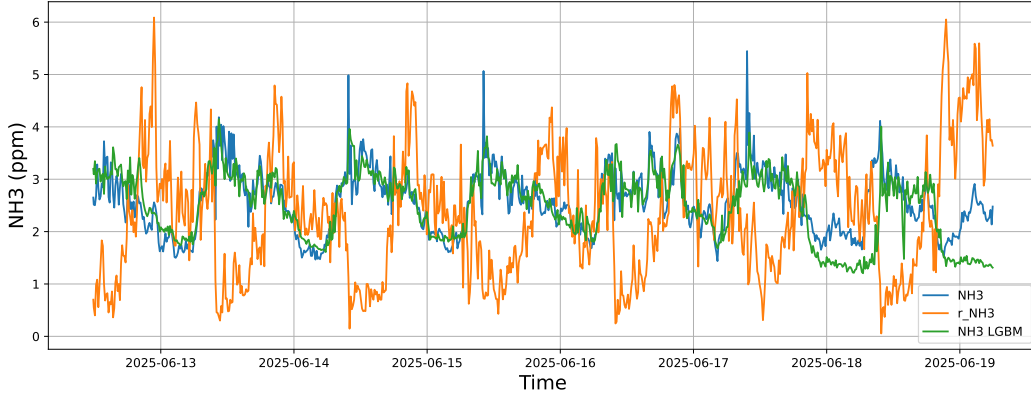


3.1.1.3 Time window : ALL

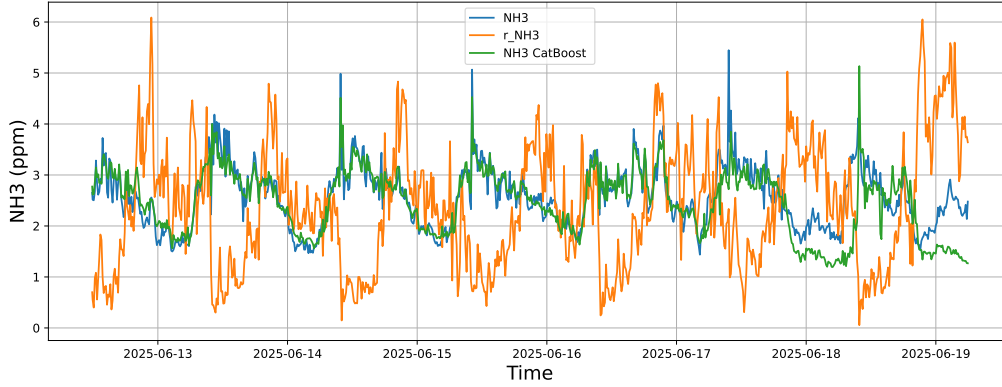
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



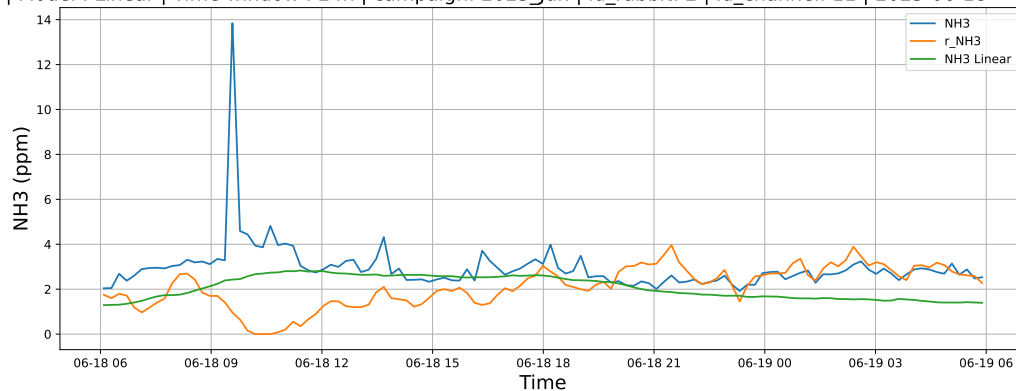
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



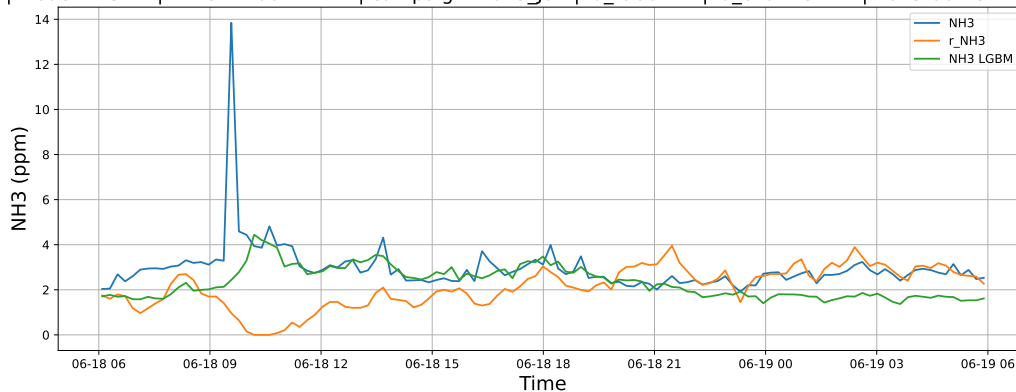
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

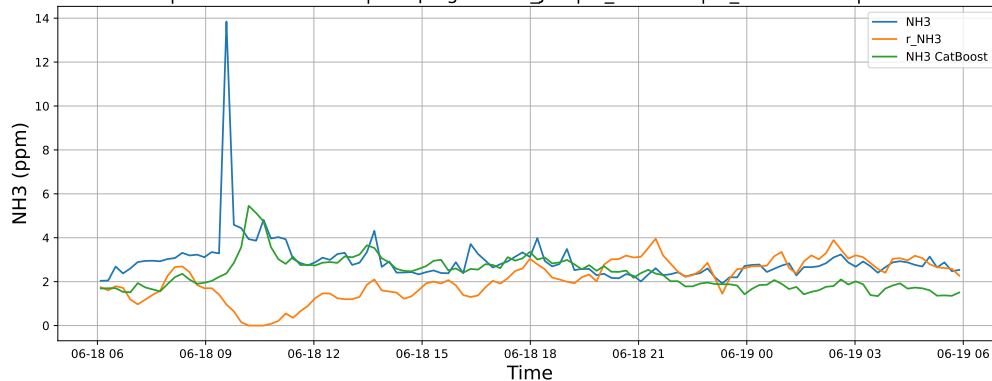
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

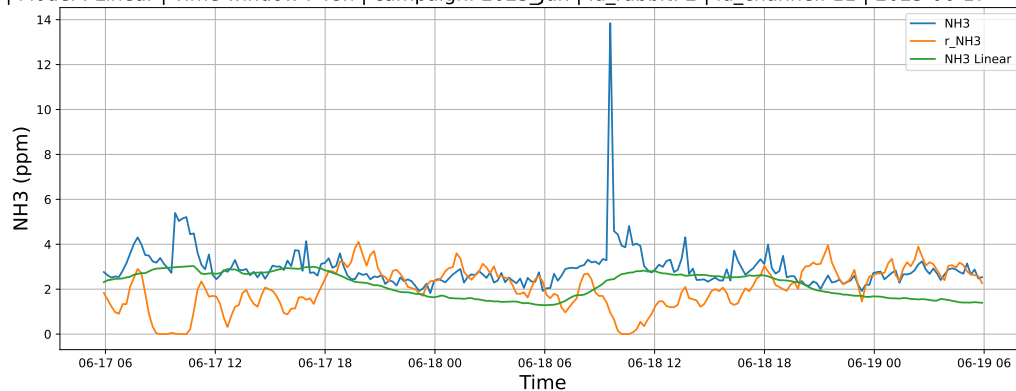


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

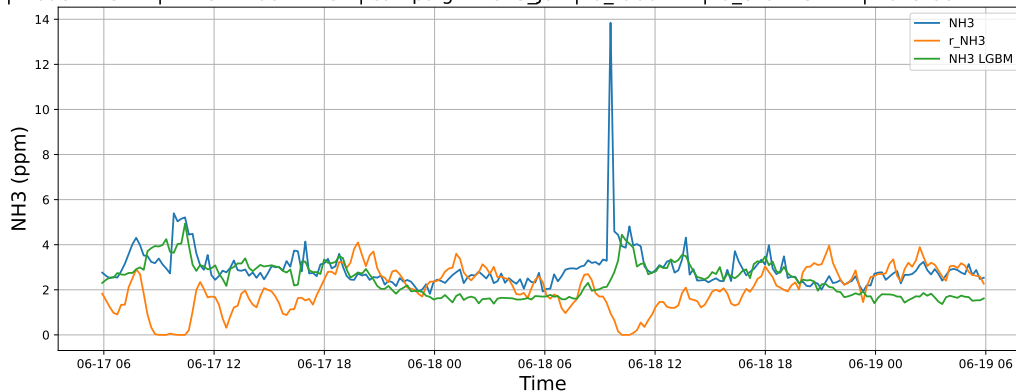


3.1.2.2 Time window : 48

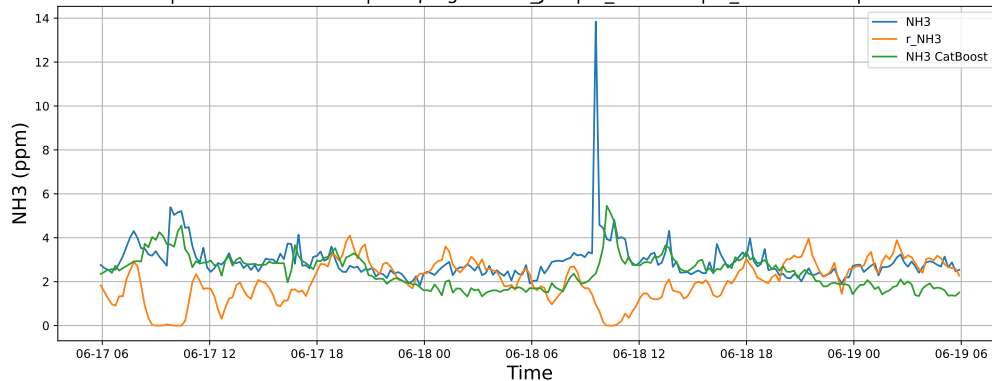
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

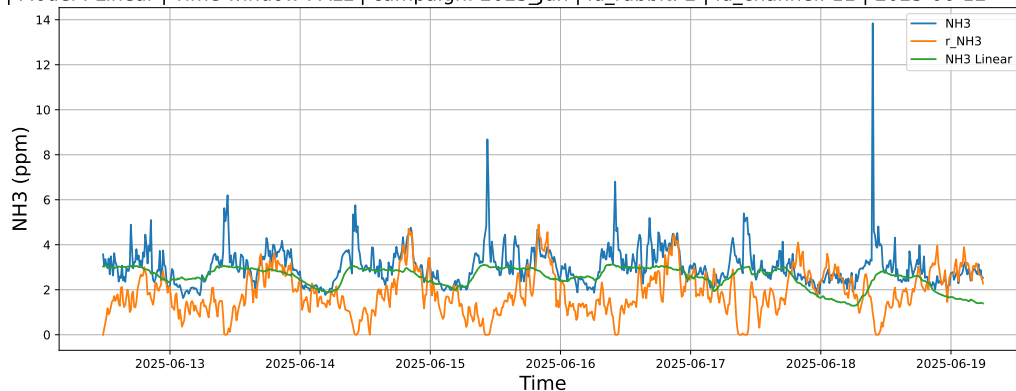


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

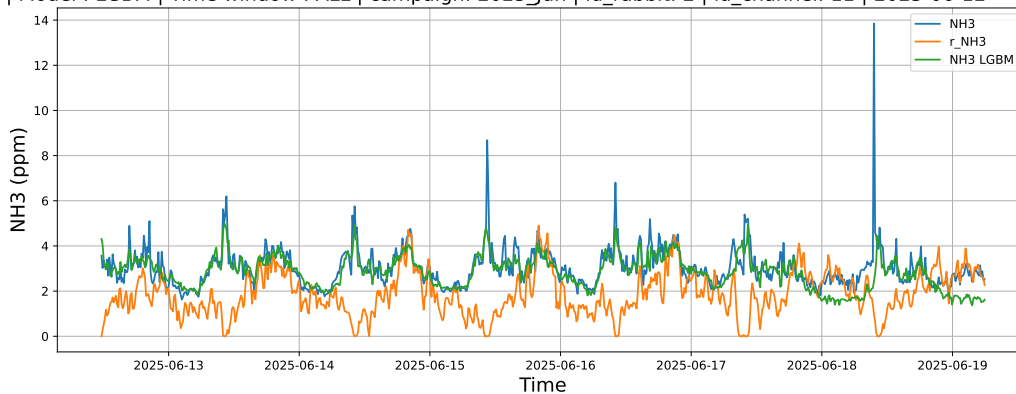


3.1.2.3 Time window : ALL

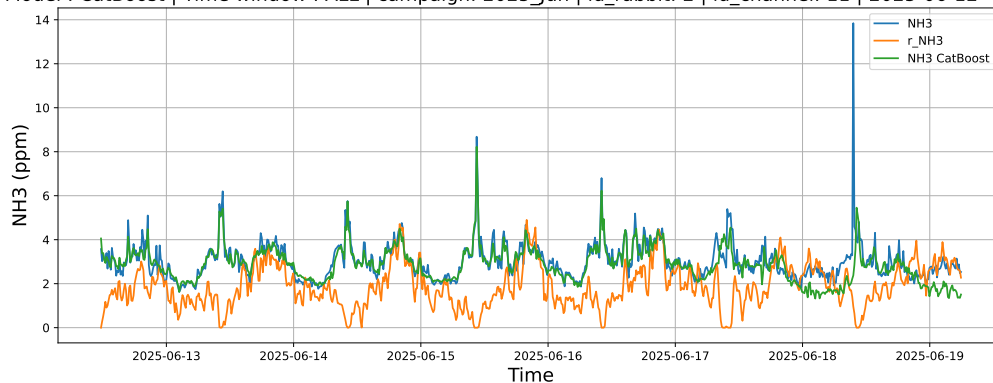
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



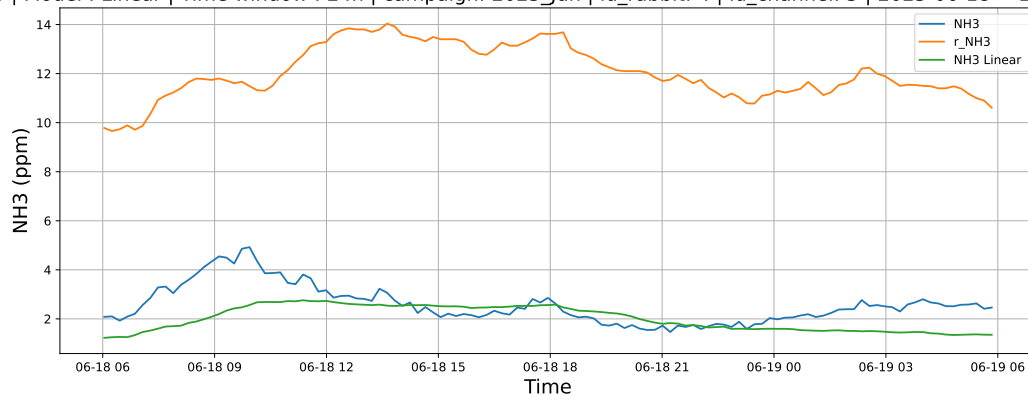
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



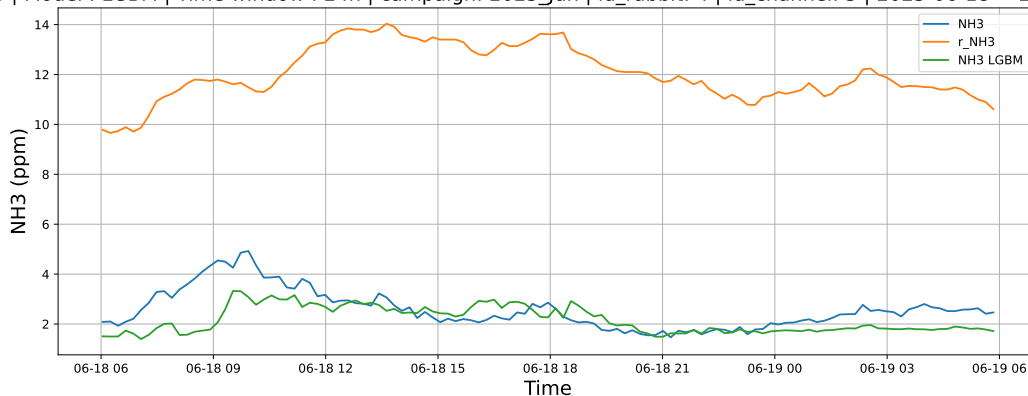
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

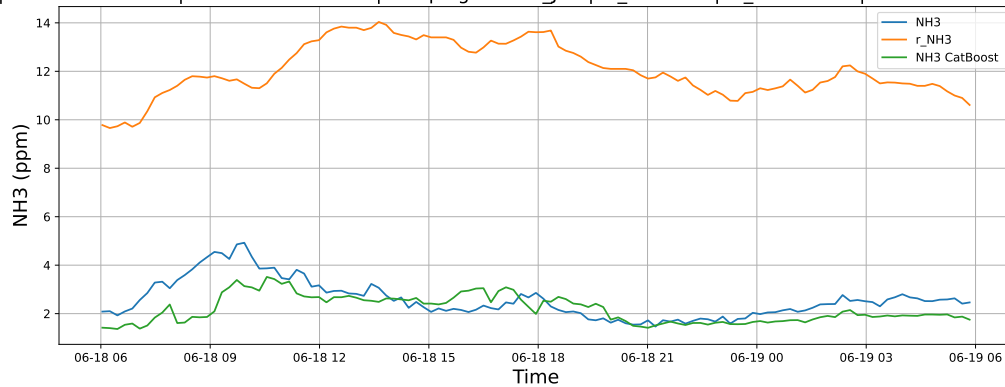
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

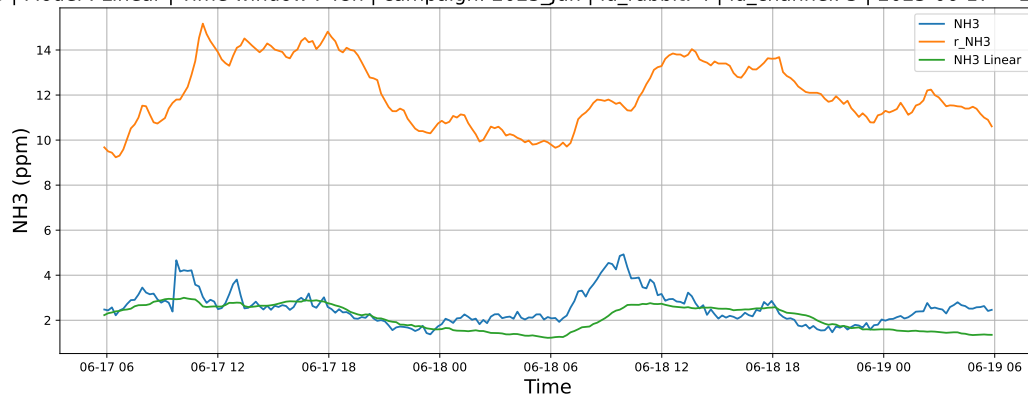


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

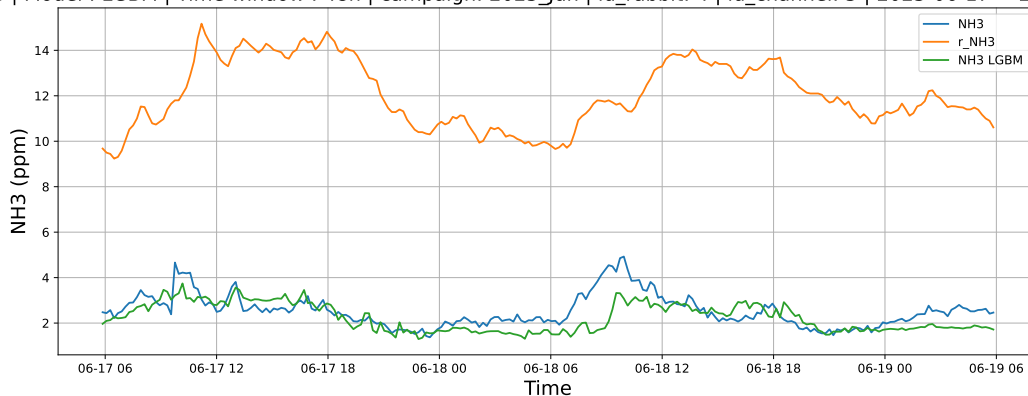


3.1.3.2 Time window : 48

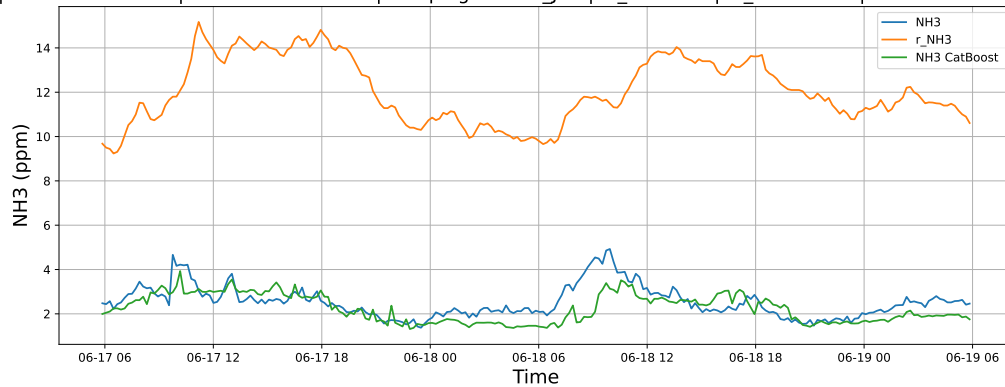
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

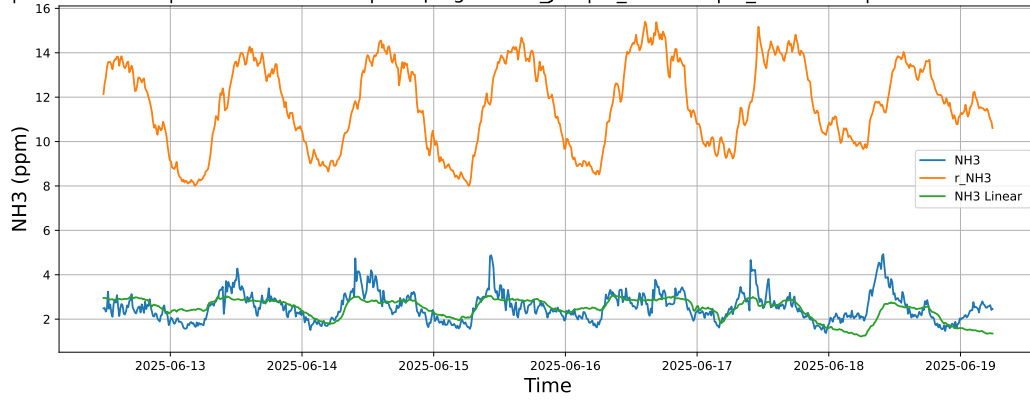


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

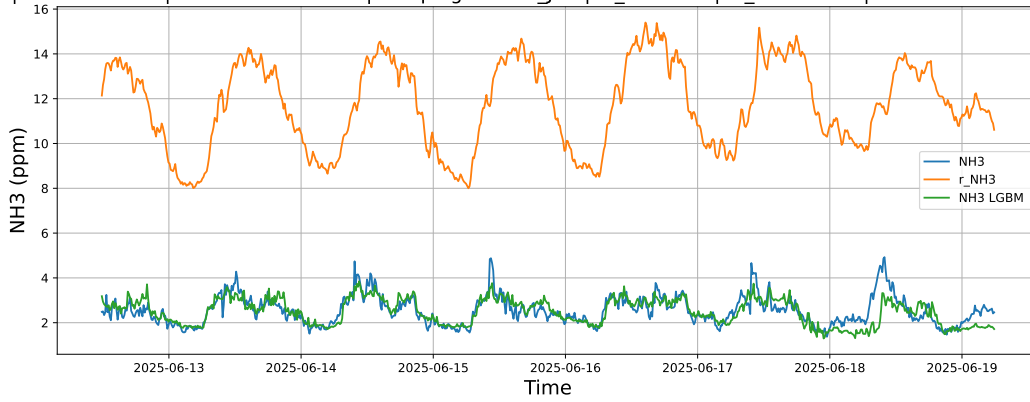


3.1.3.3 Time window : ALL

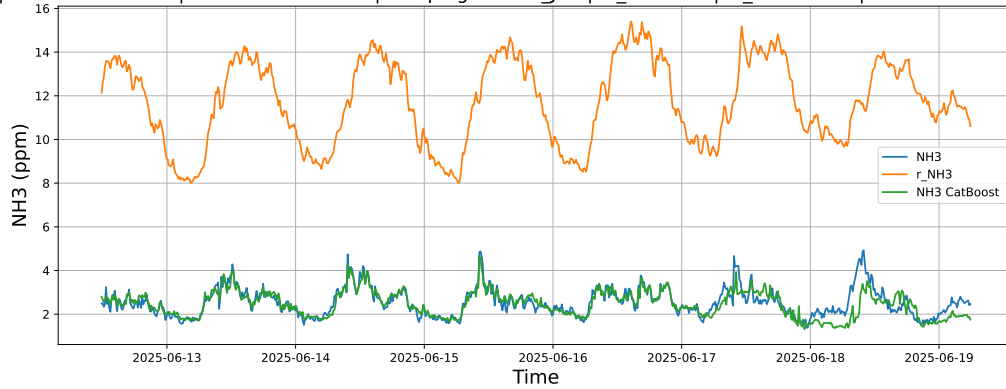
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



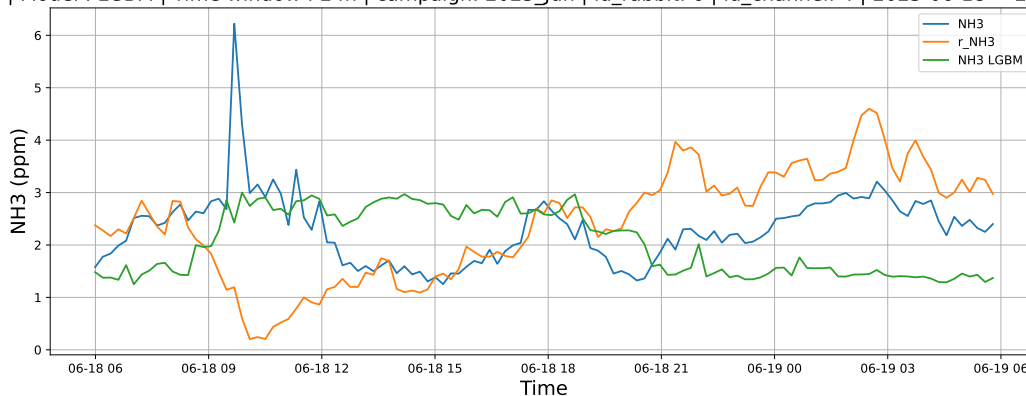
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

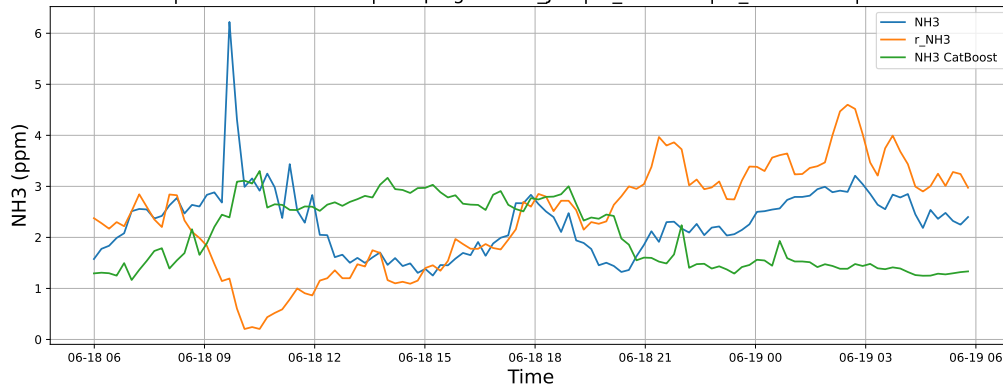
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

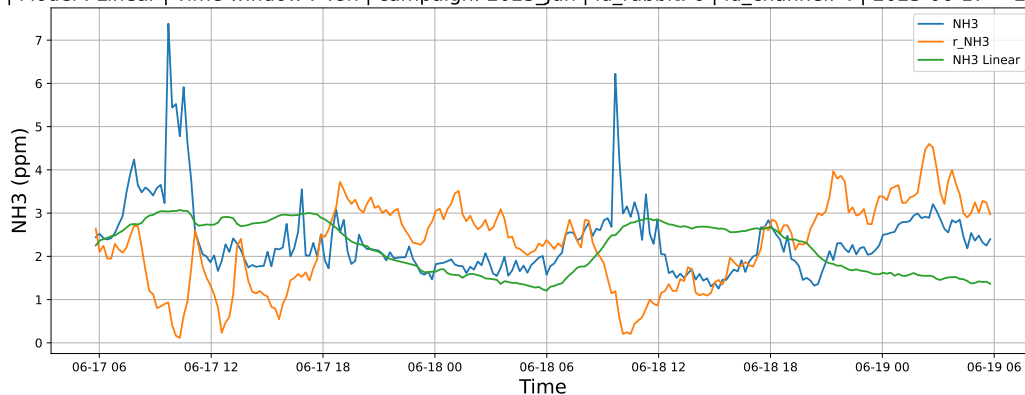


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

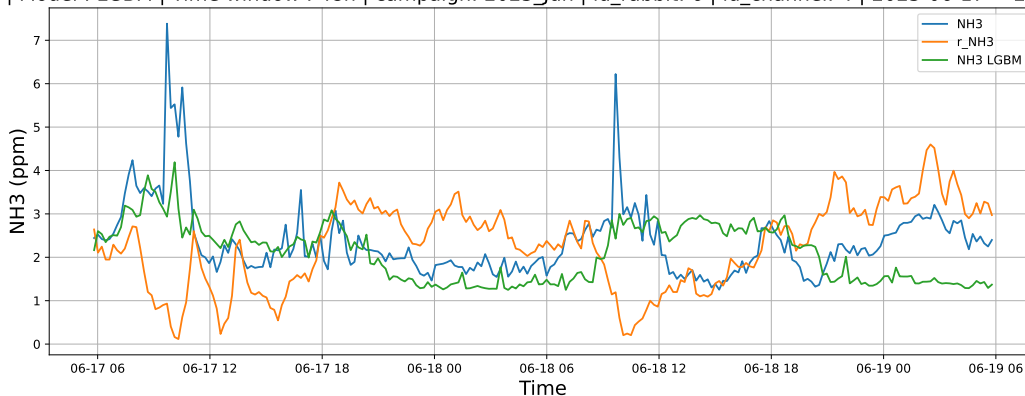


3.1.4.2 Time window : 48

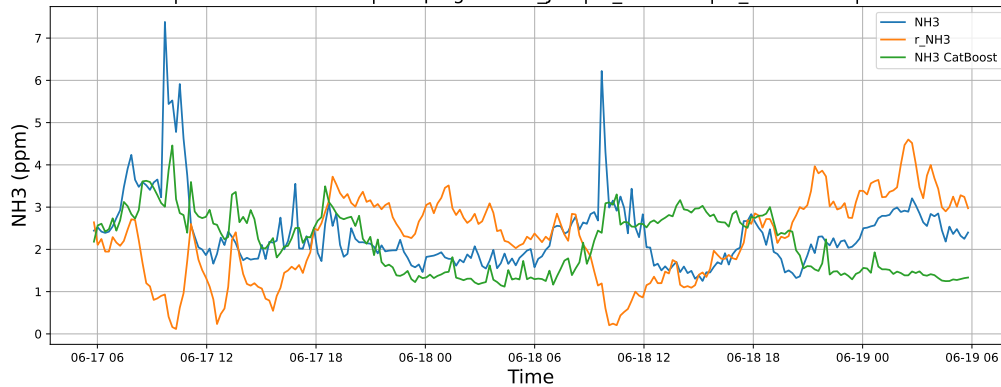
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

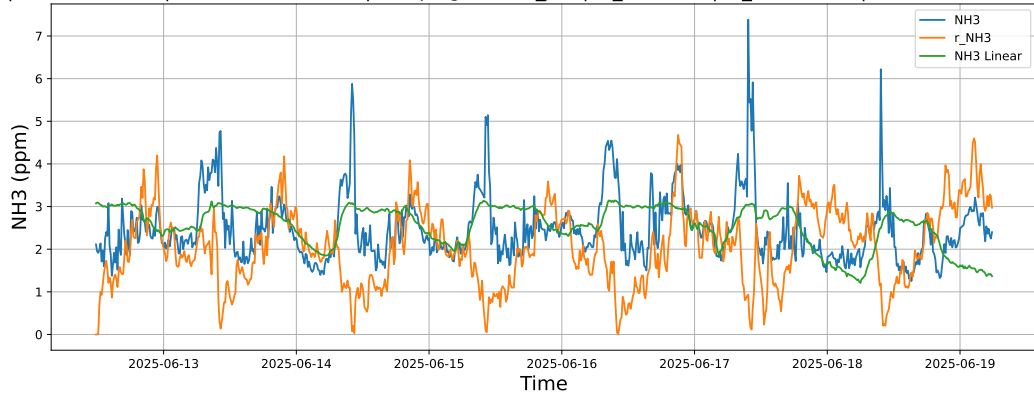


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

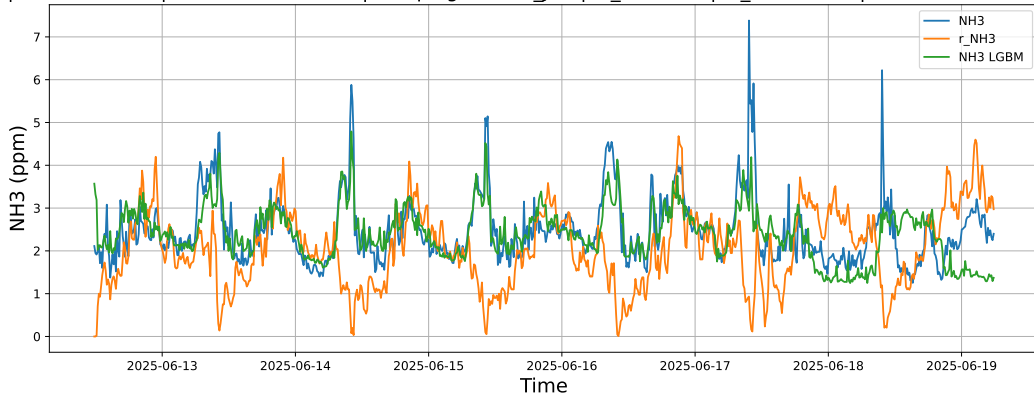


3.1.4.3 Time window : ALL

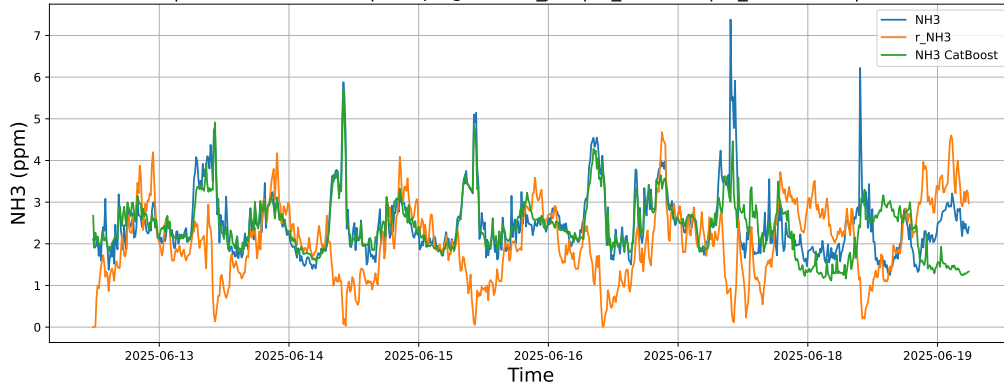
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



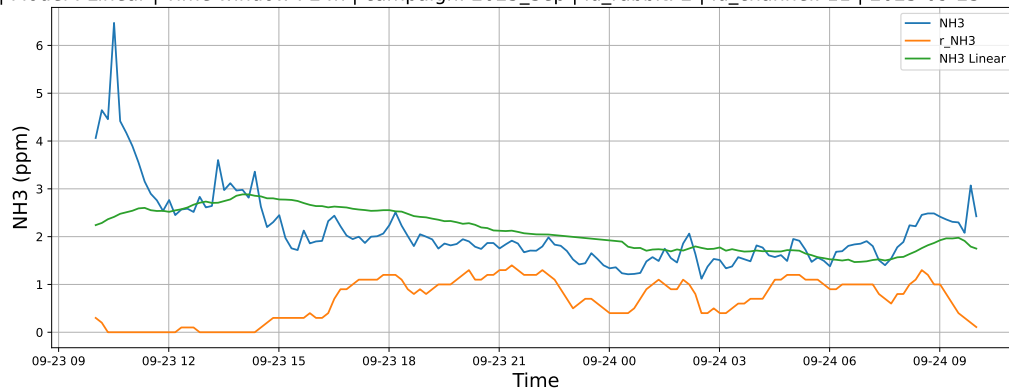
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



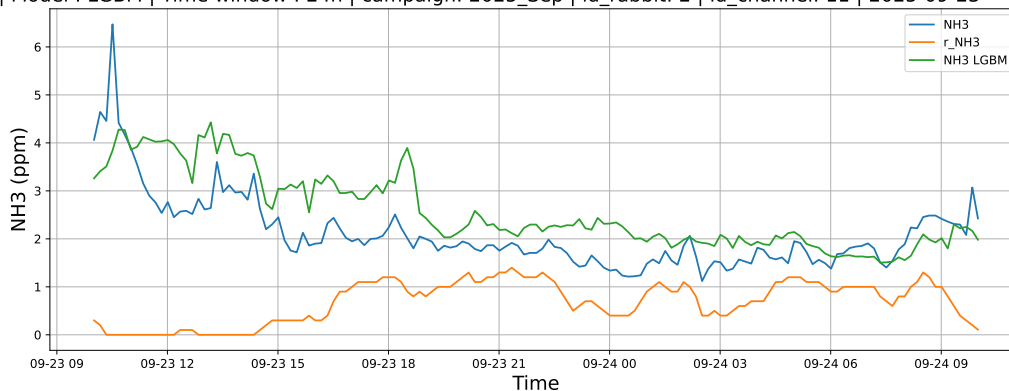
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

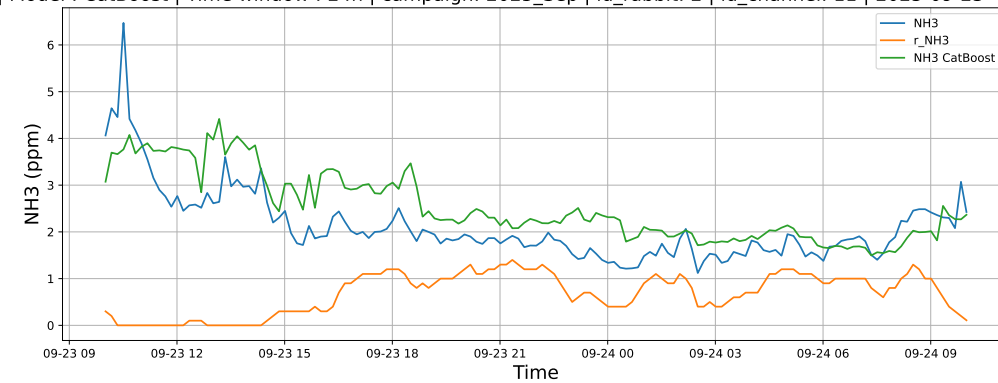
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

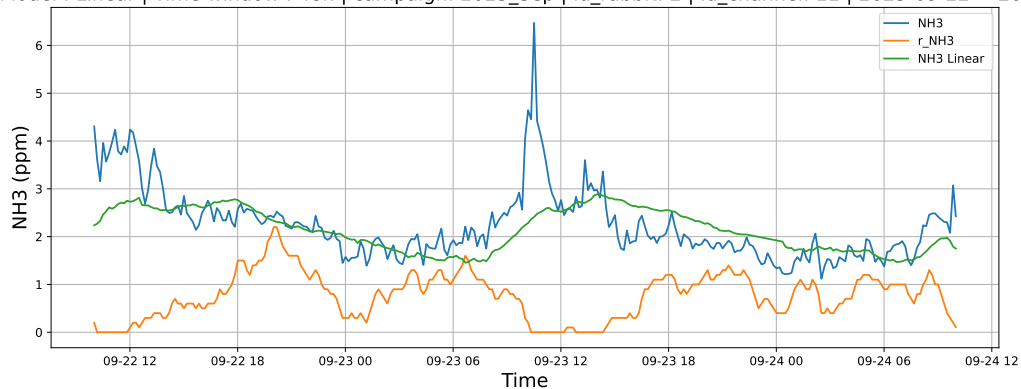


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

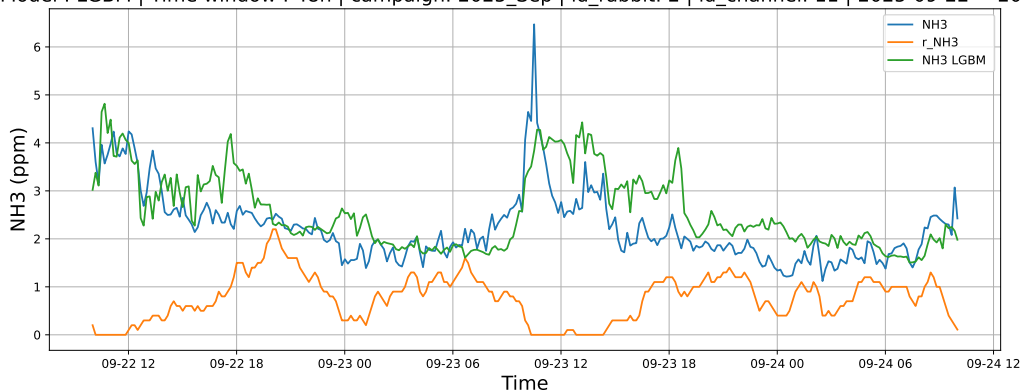


3.1.5.2 Time window : 48

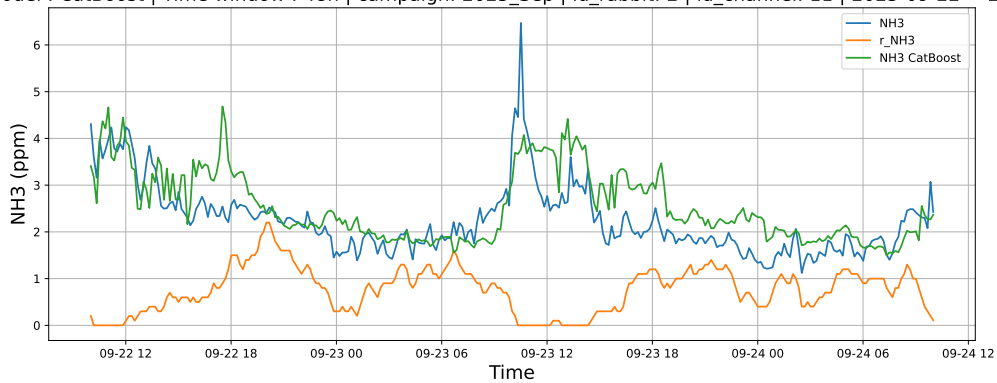
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

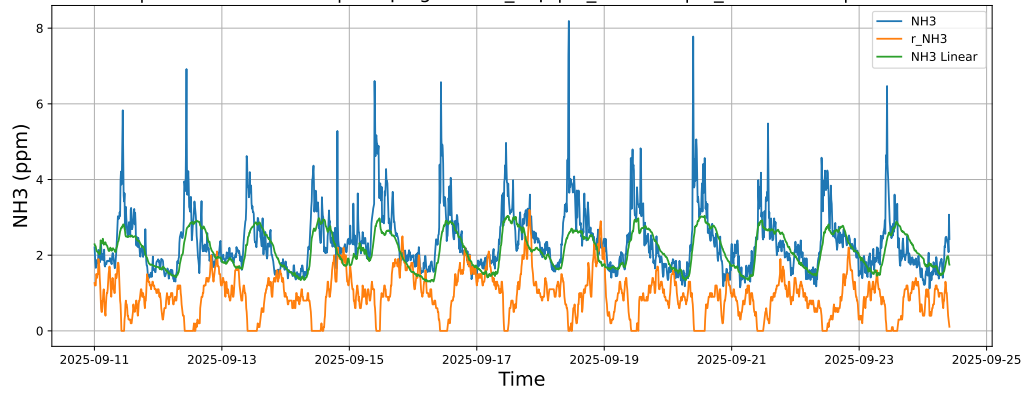


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

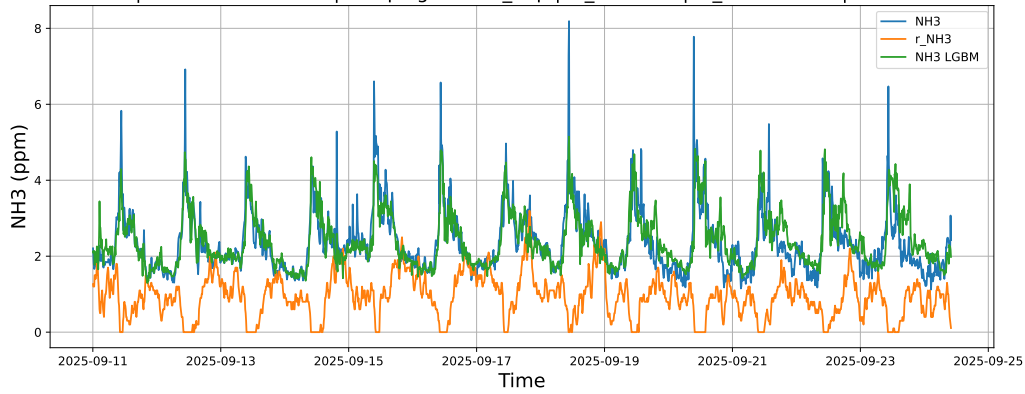


3.1.5.3 Time window : ALL

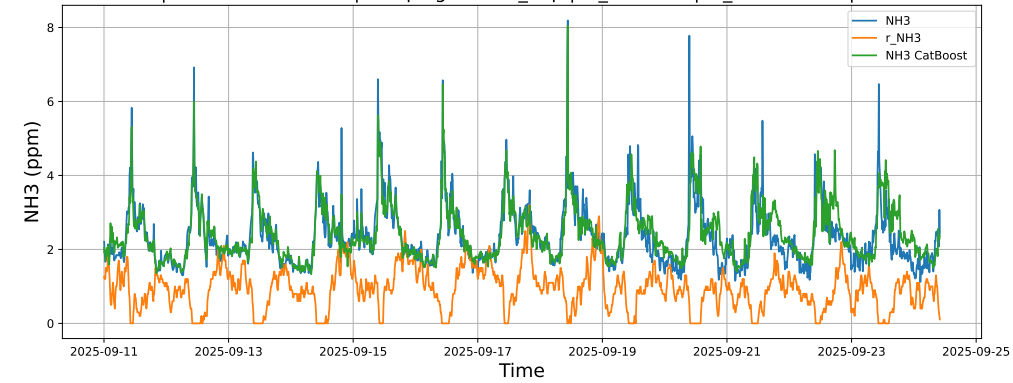
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



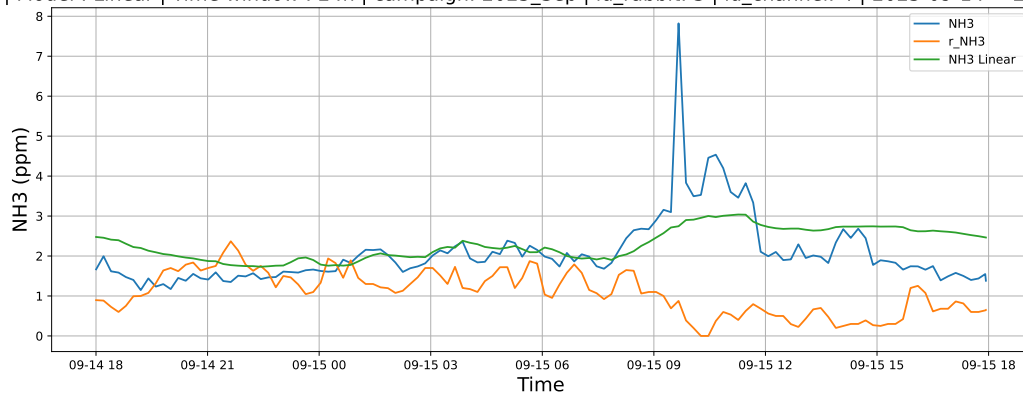
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



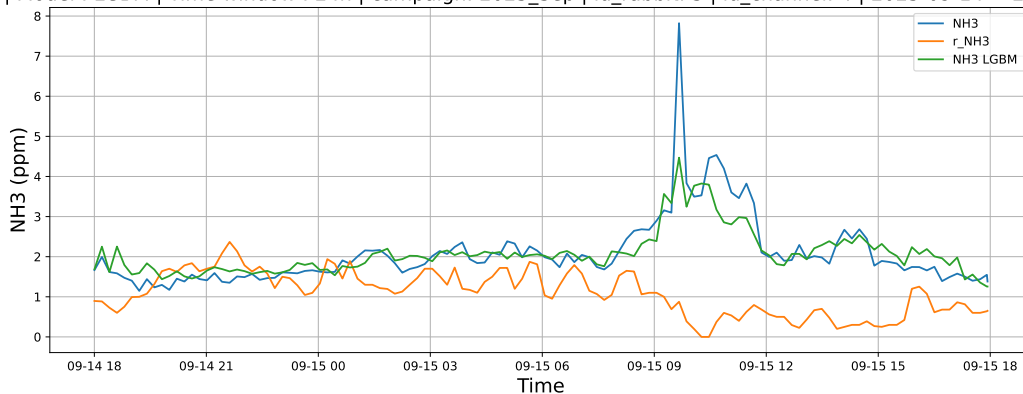
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

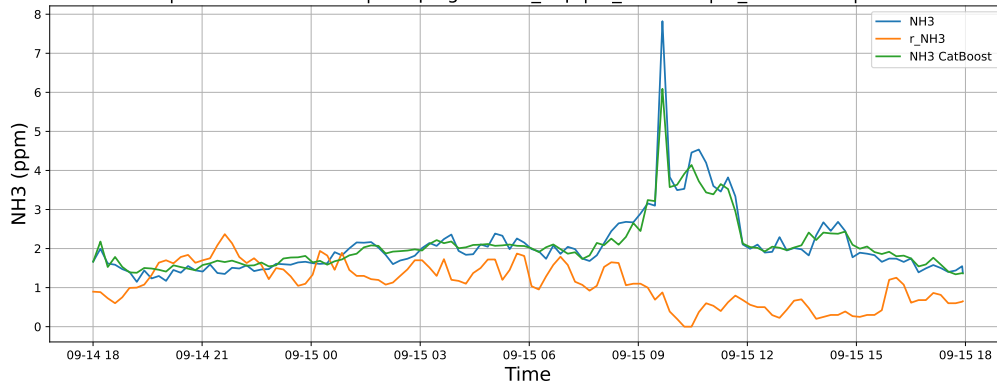
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

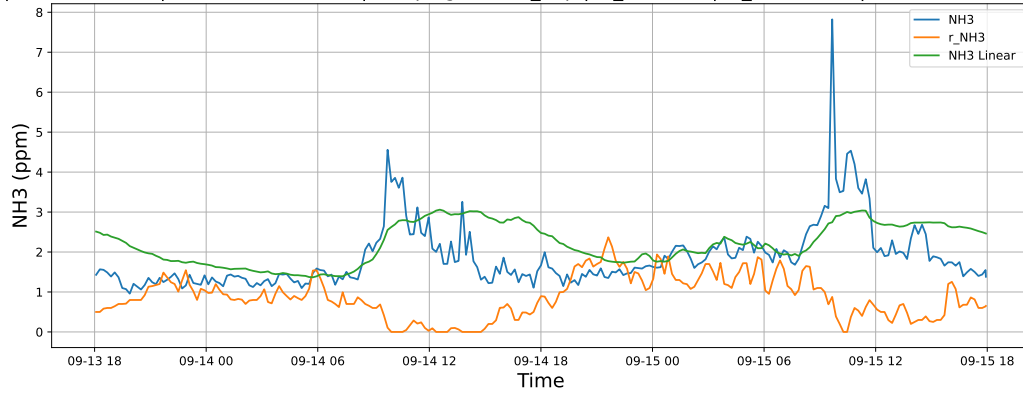


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

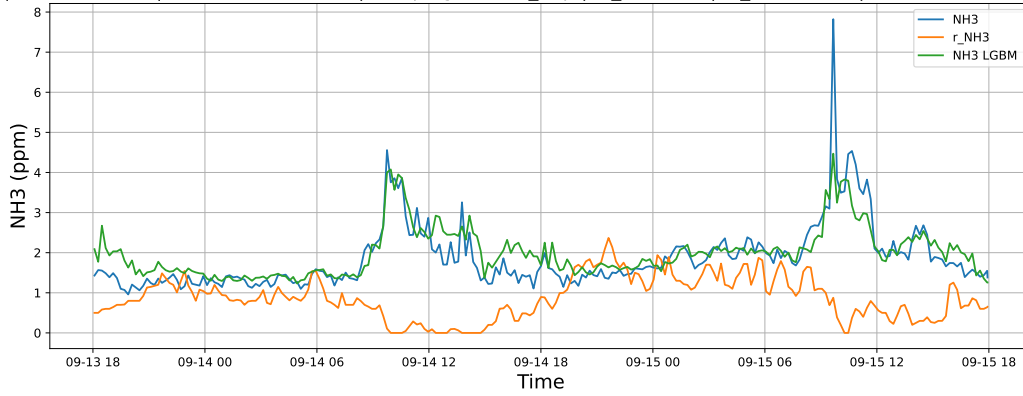


3.1.6.2 Time window : 48

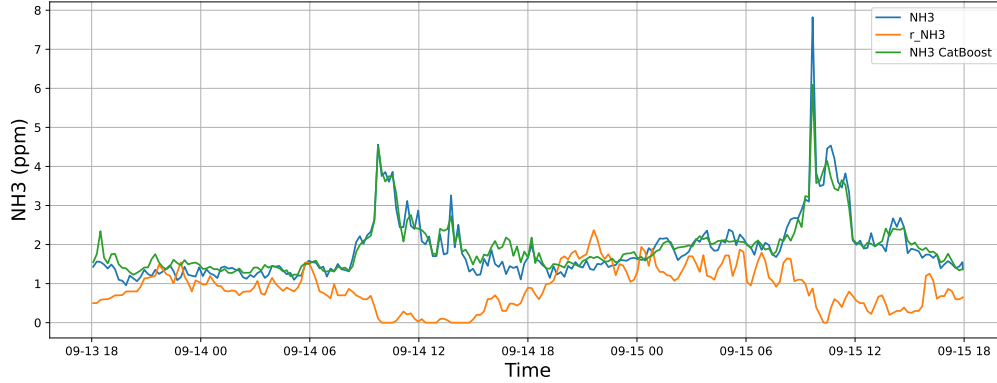
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

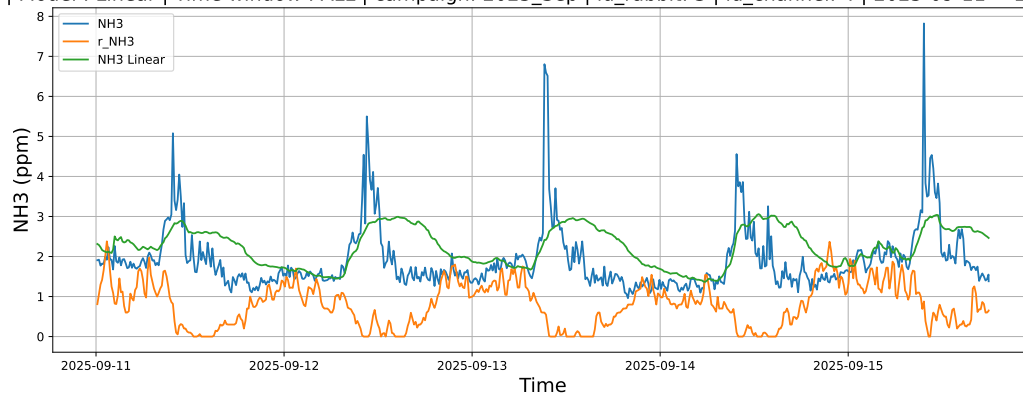


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

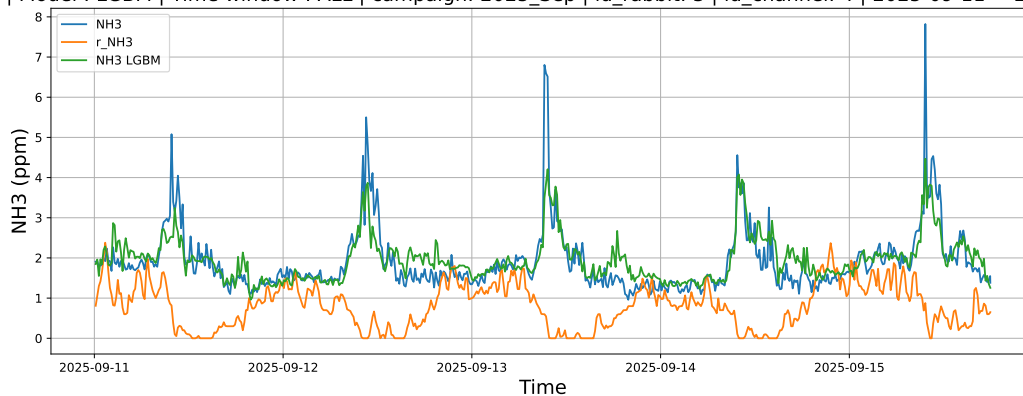


3.1.6.3 Time window : ALL

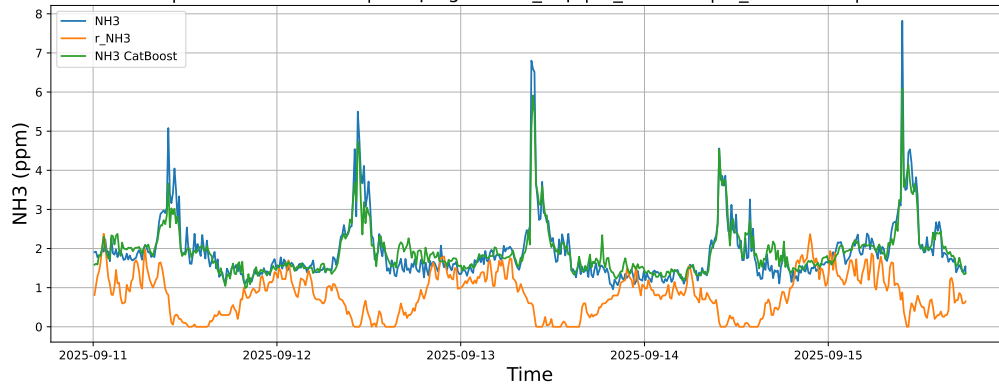
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



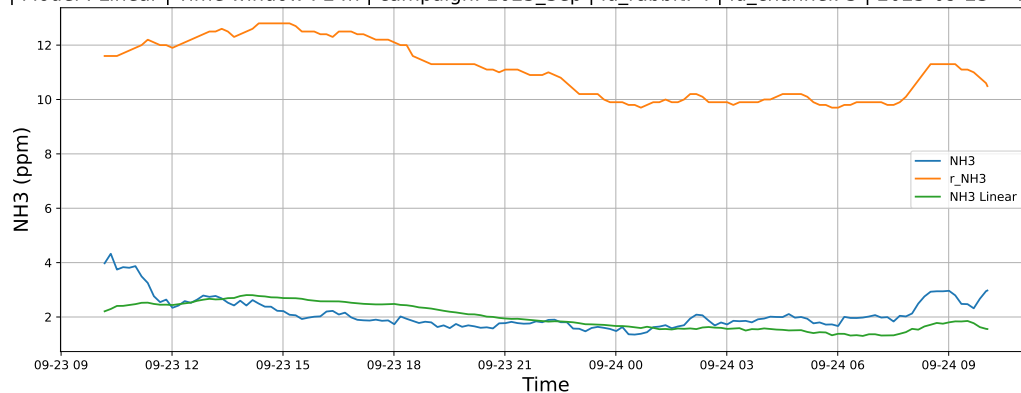
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



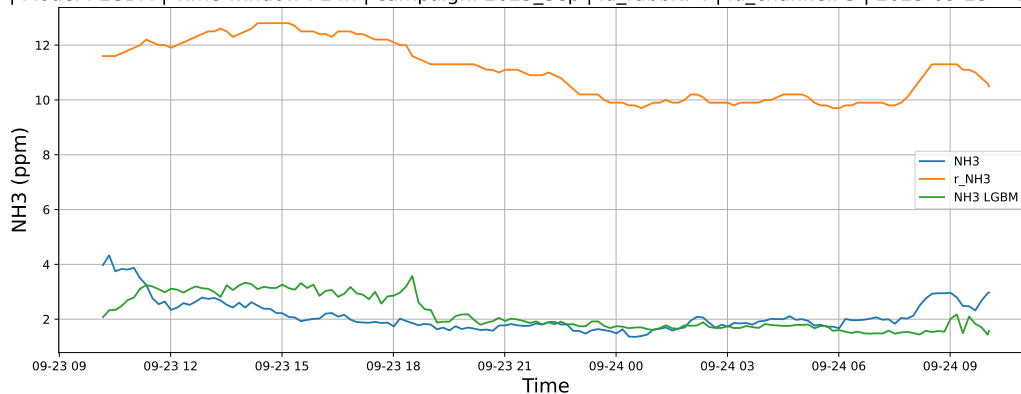
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

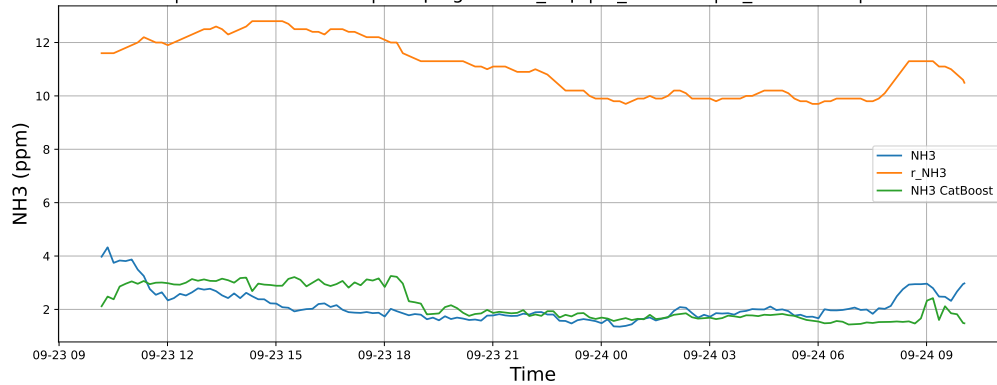
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

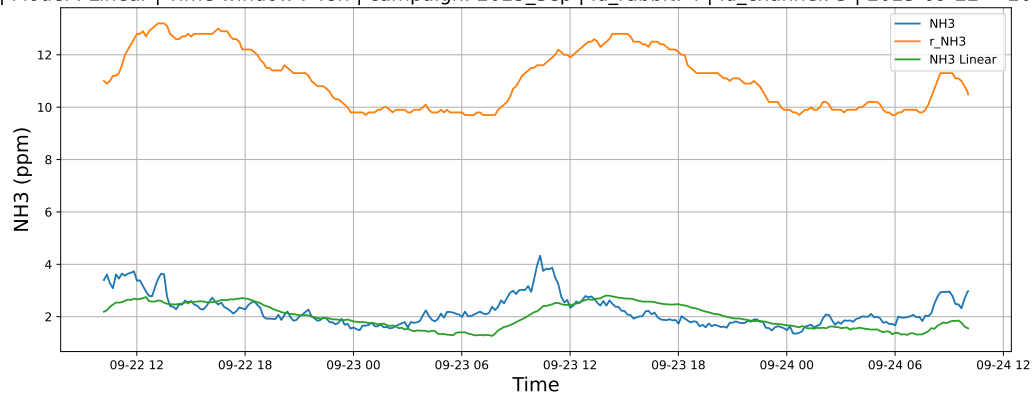


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

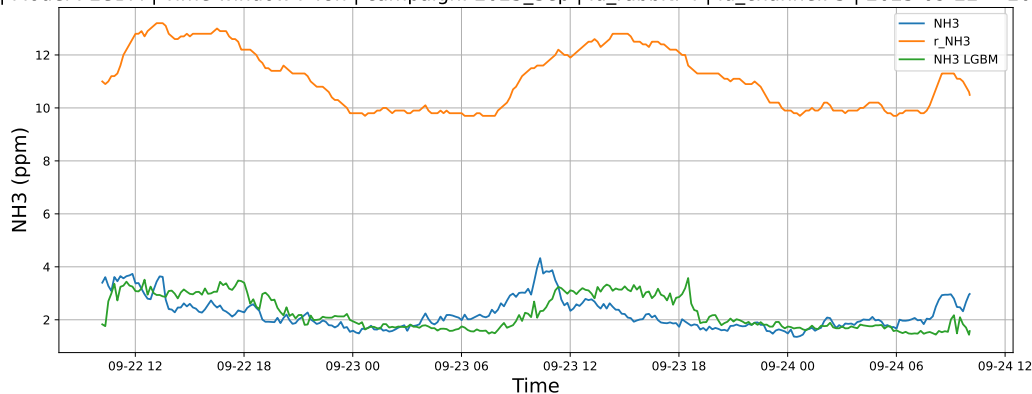


3.1.7.2 Time window : 48

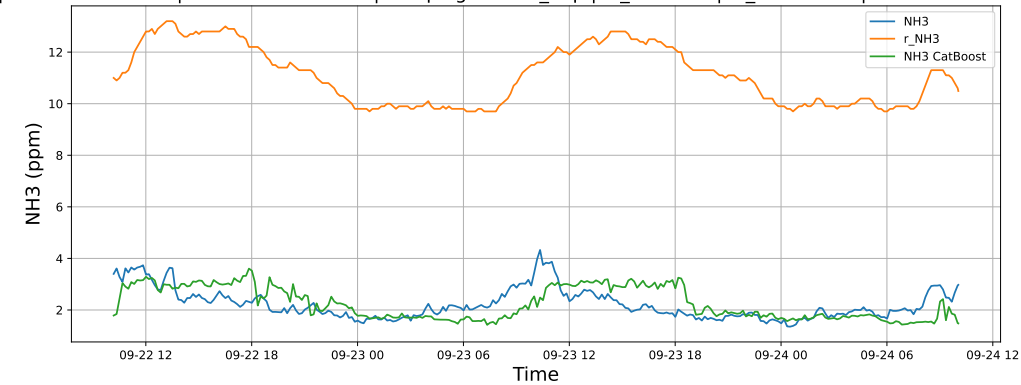
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

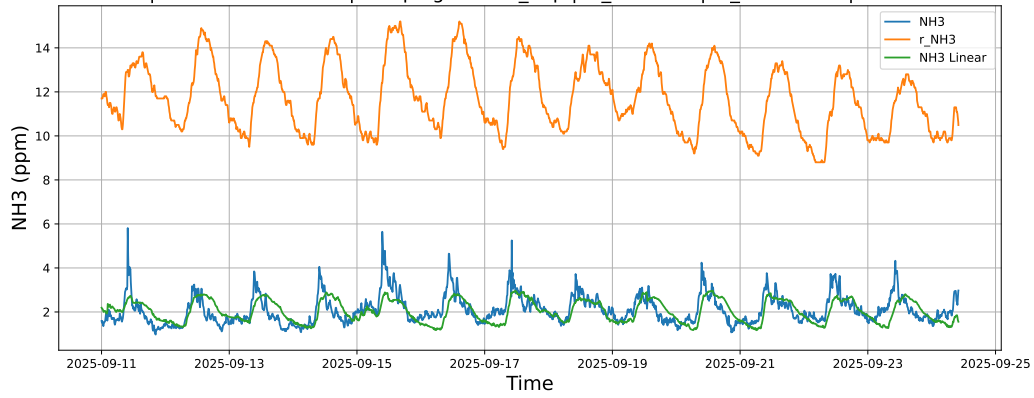


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

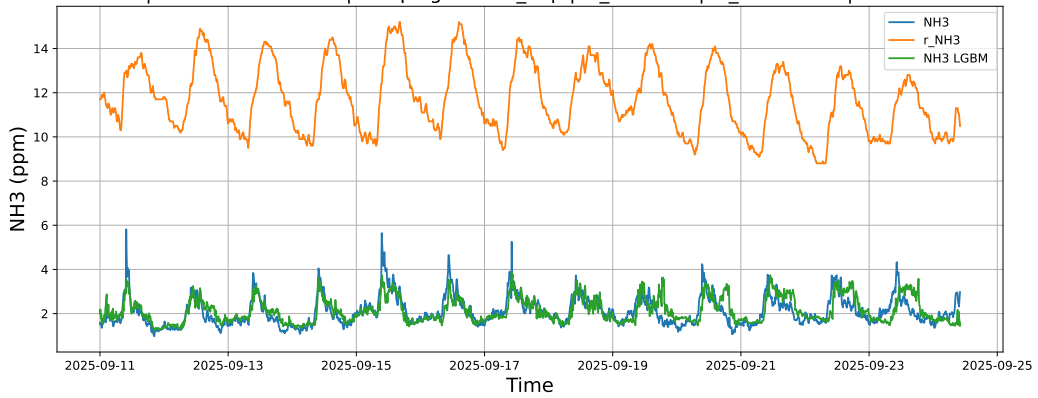


3.1.7.3 Time window : ALL

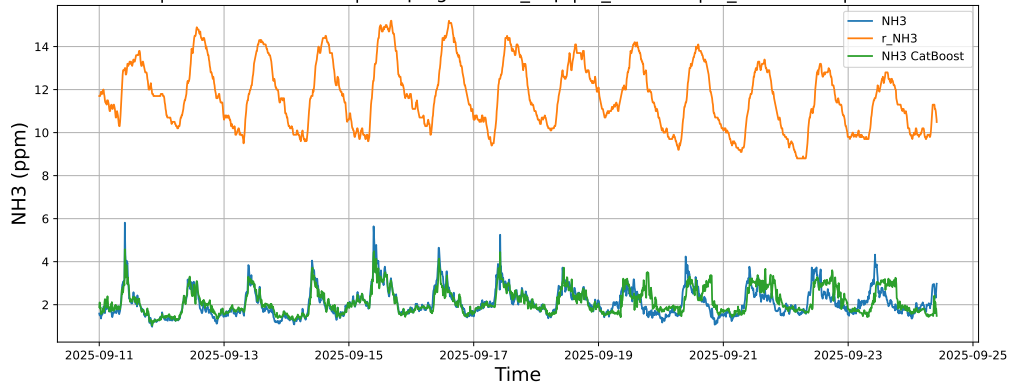
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



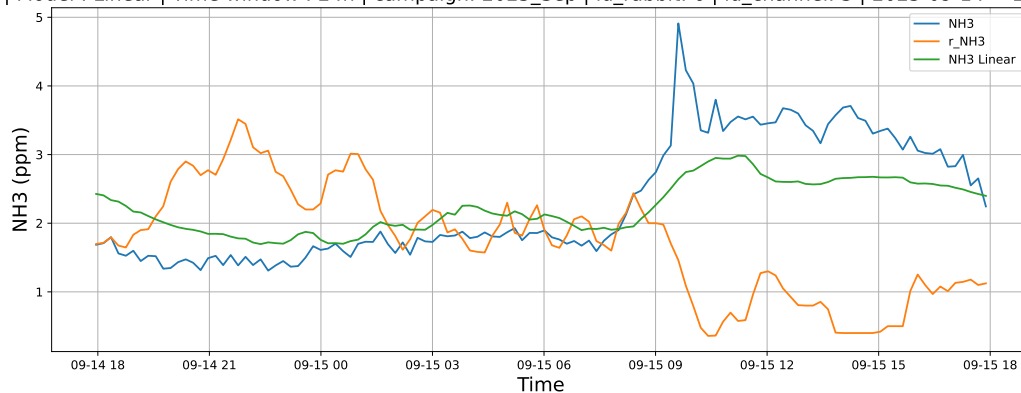
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



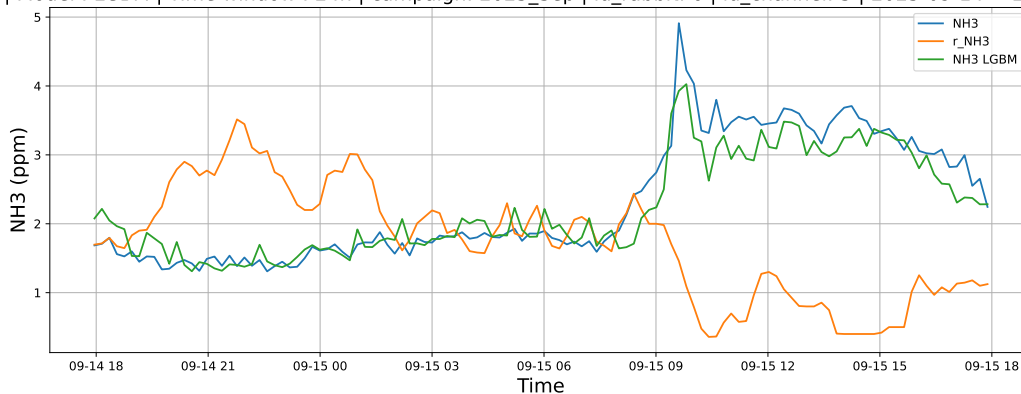
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

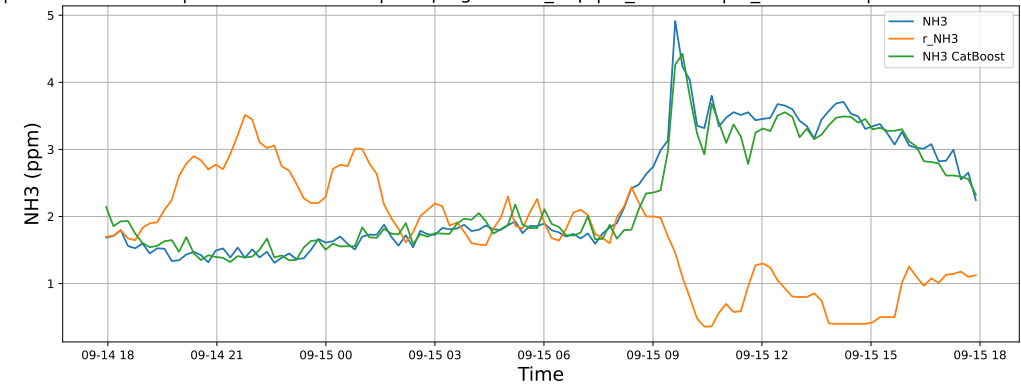
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

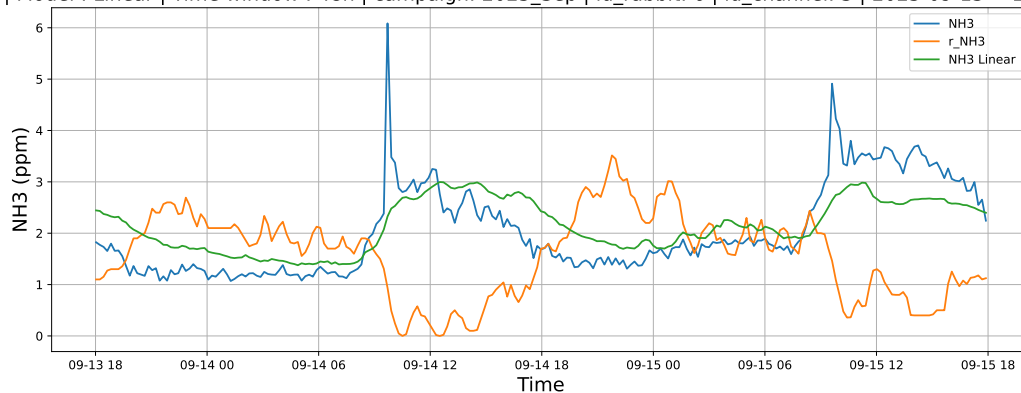


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

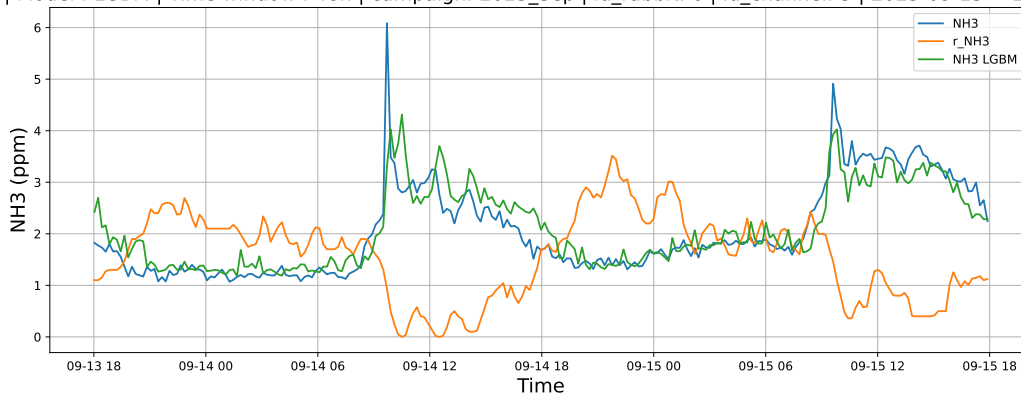


3.1.8.2 Time window : 48

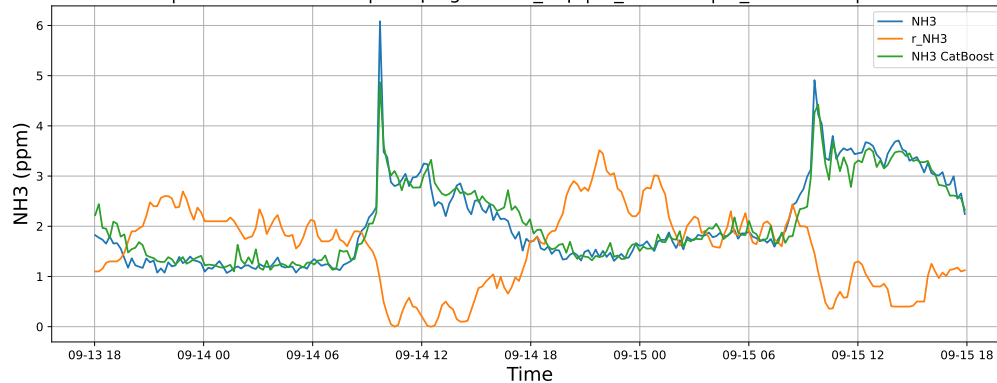
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

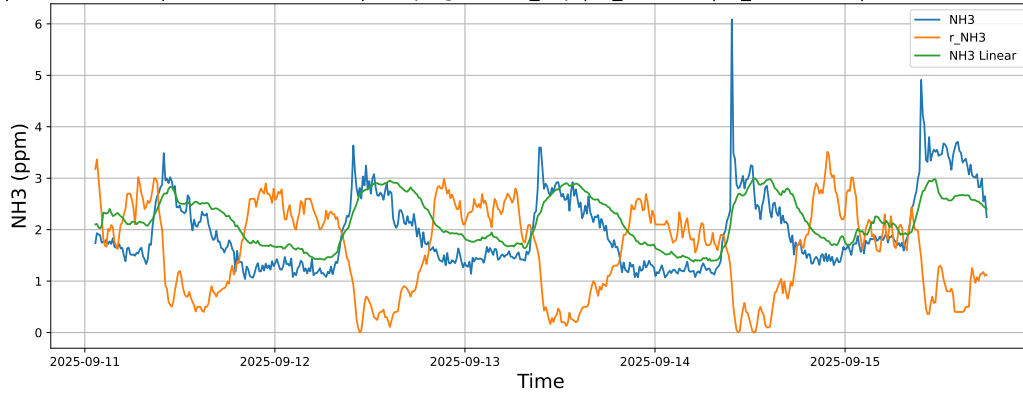


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

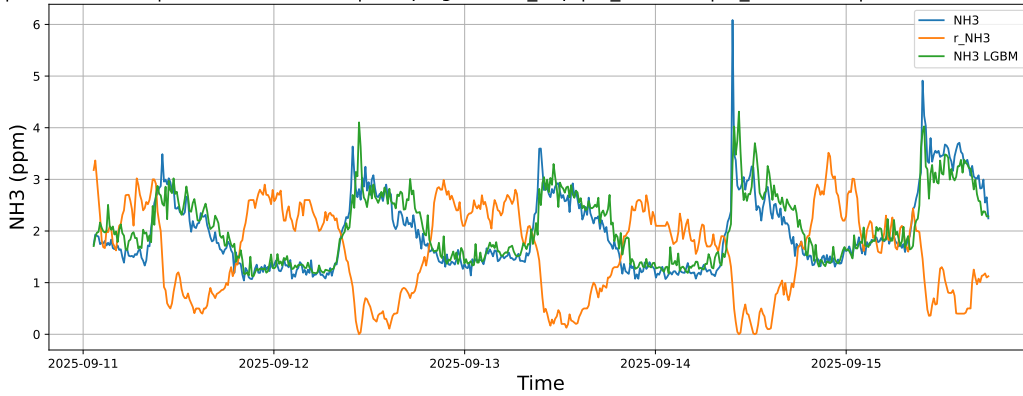


3.1.8.3 Time window : ALL

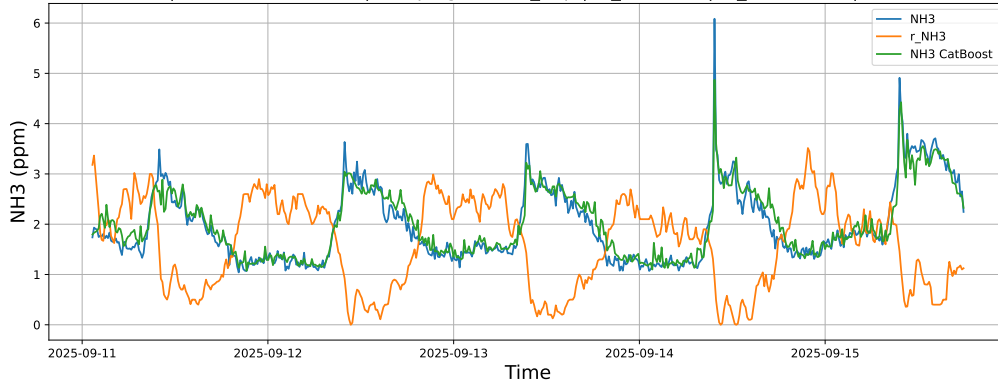
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

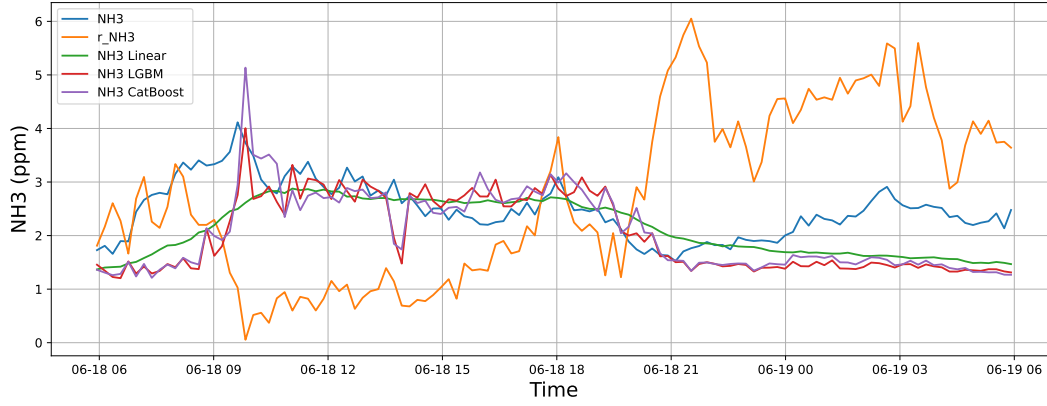


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

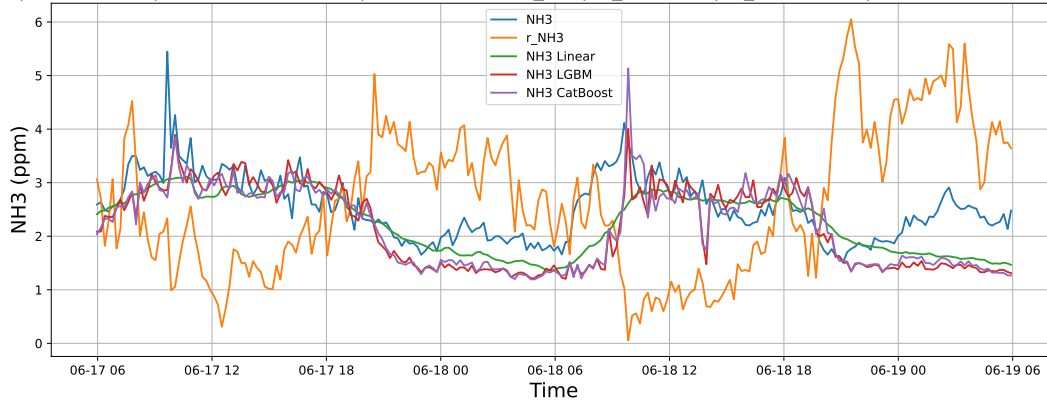
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



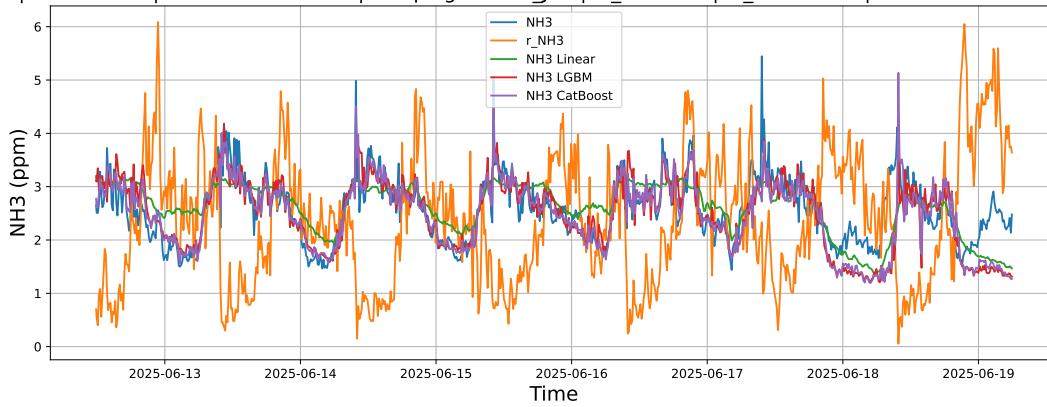
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

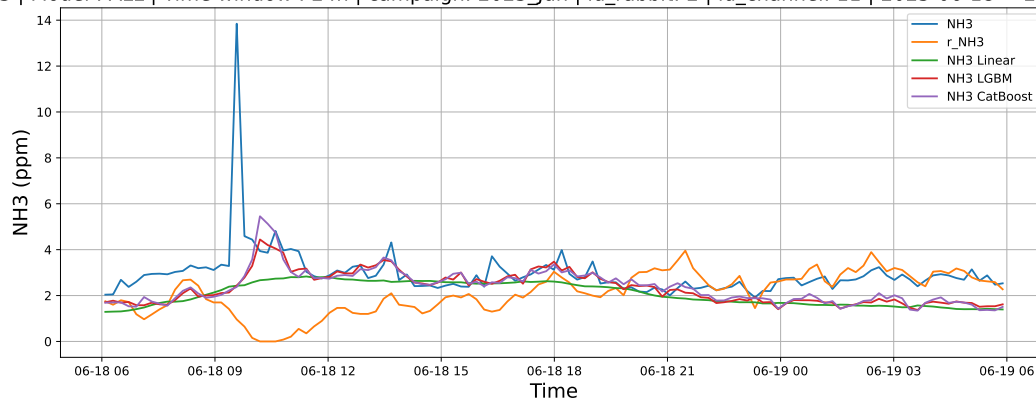
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

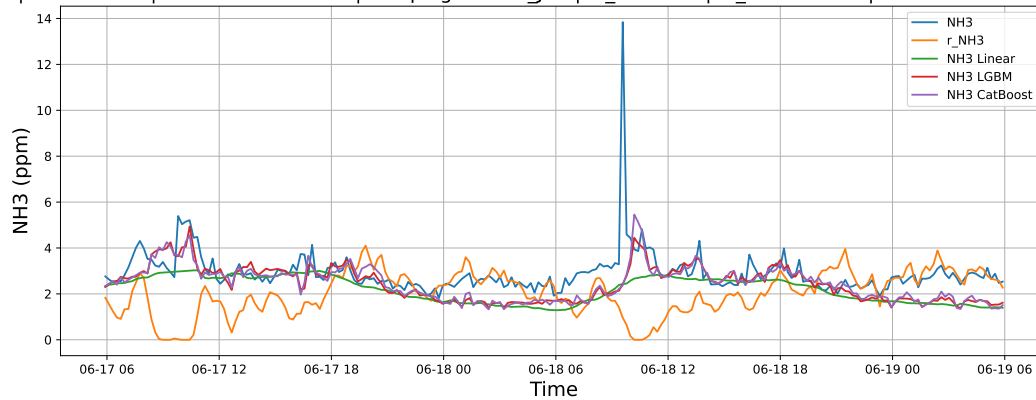
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



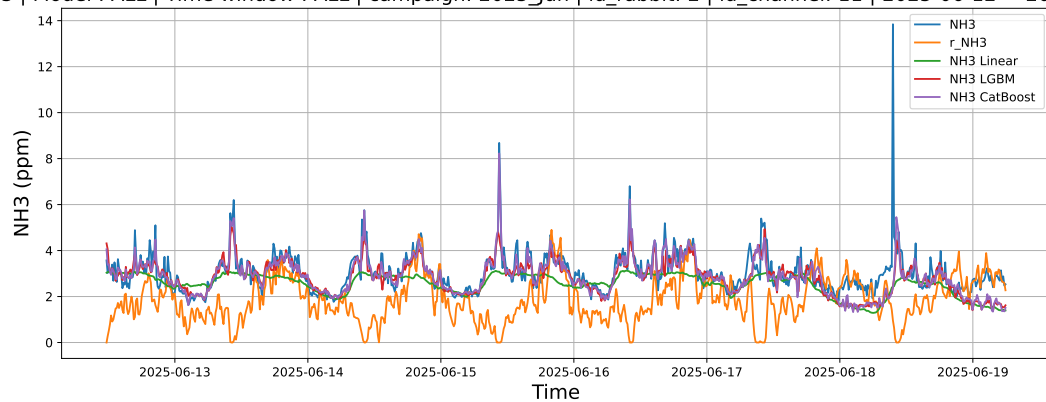
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

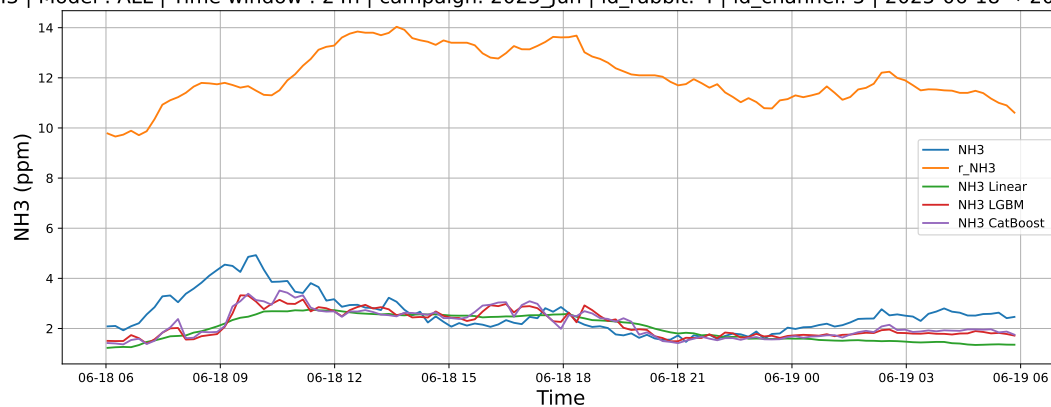
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

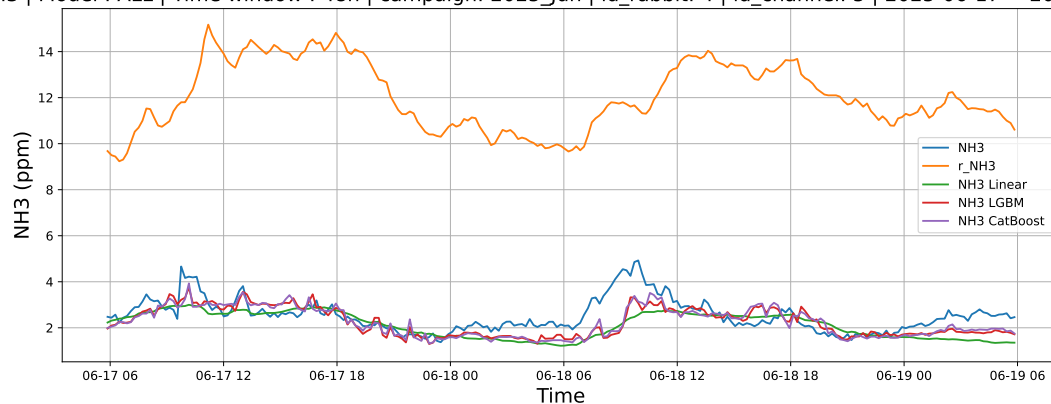
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



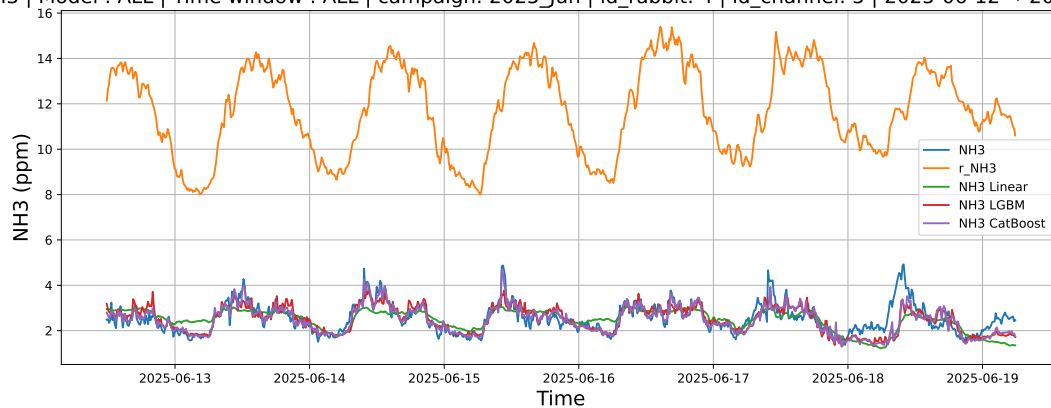
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

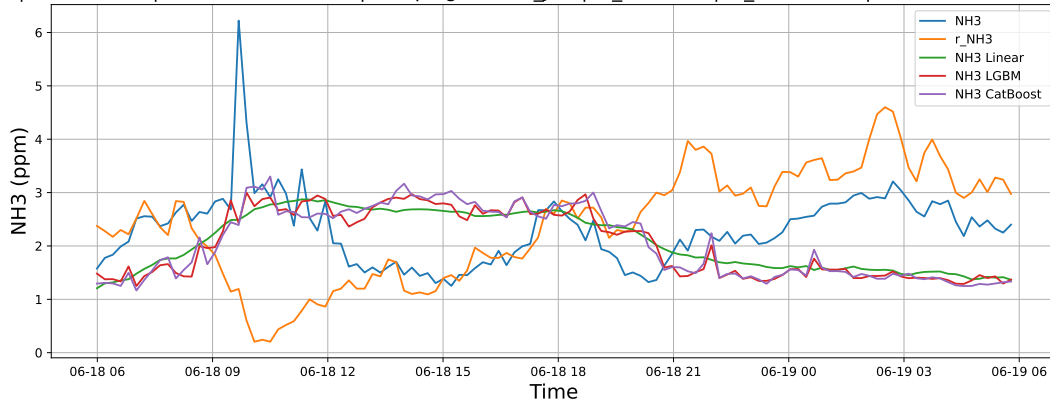
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

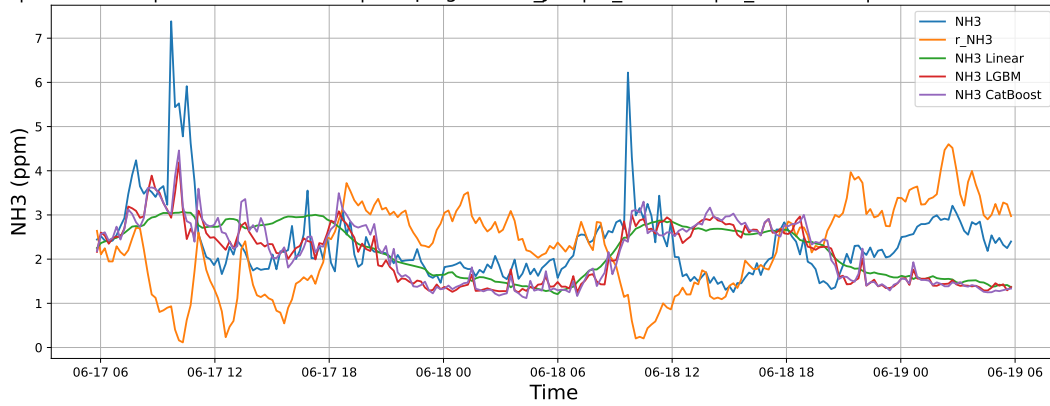
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



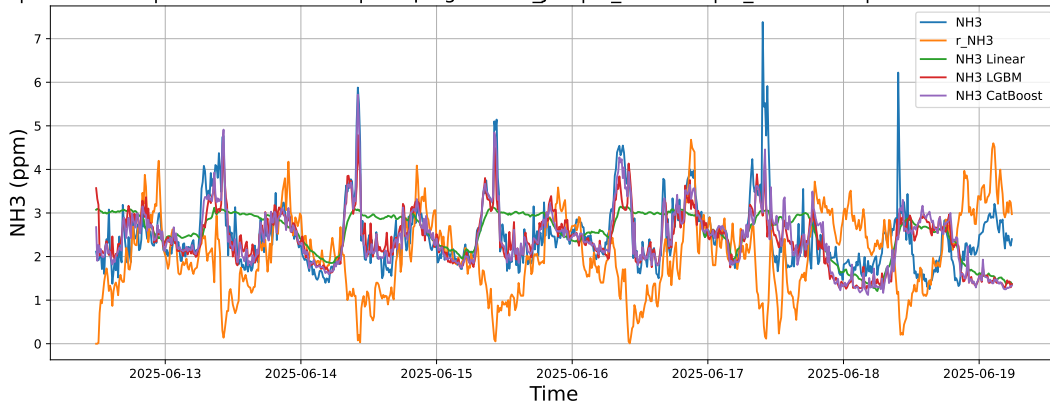
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

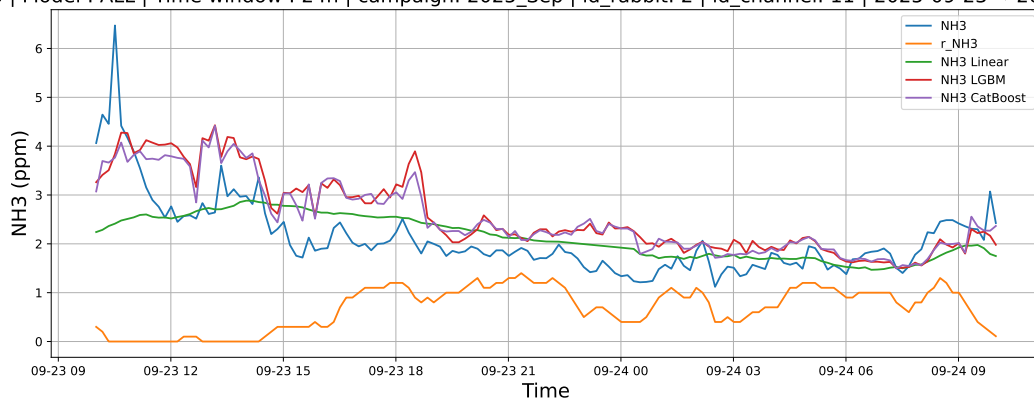
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

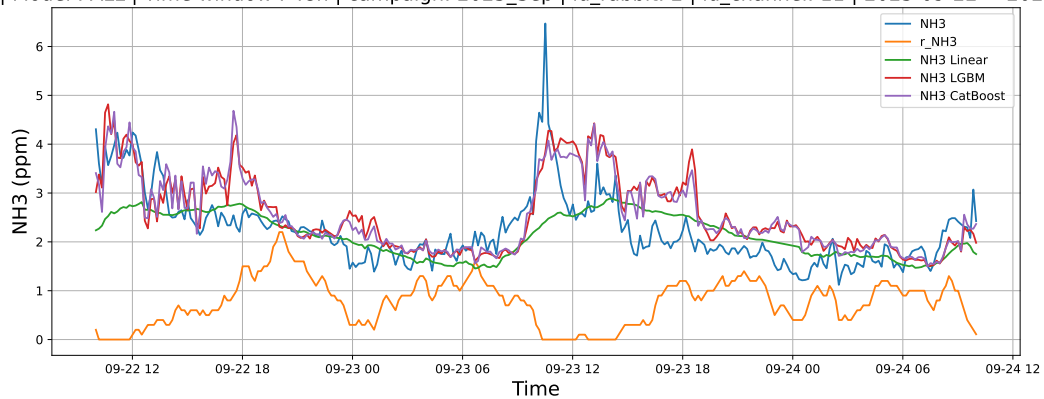
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



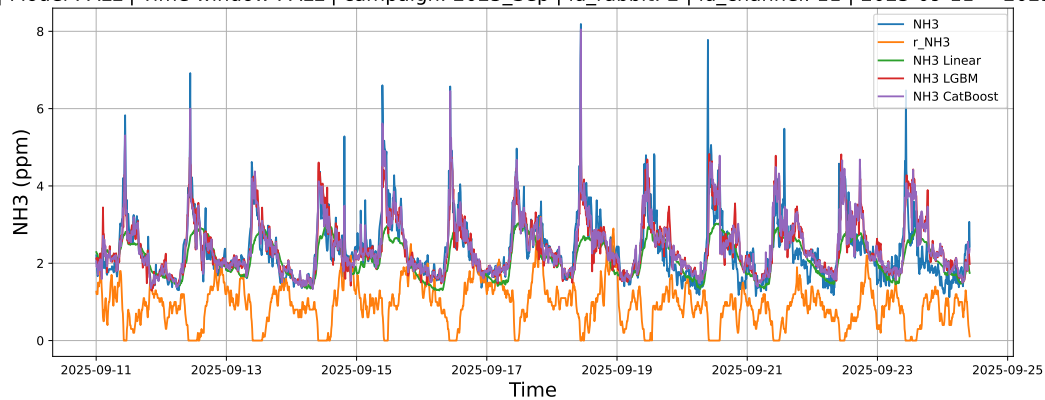
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

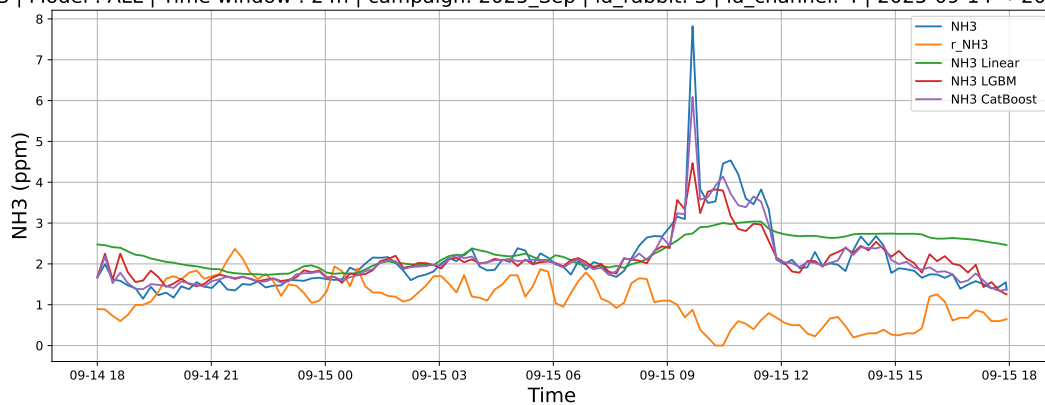
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

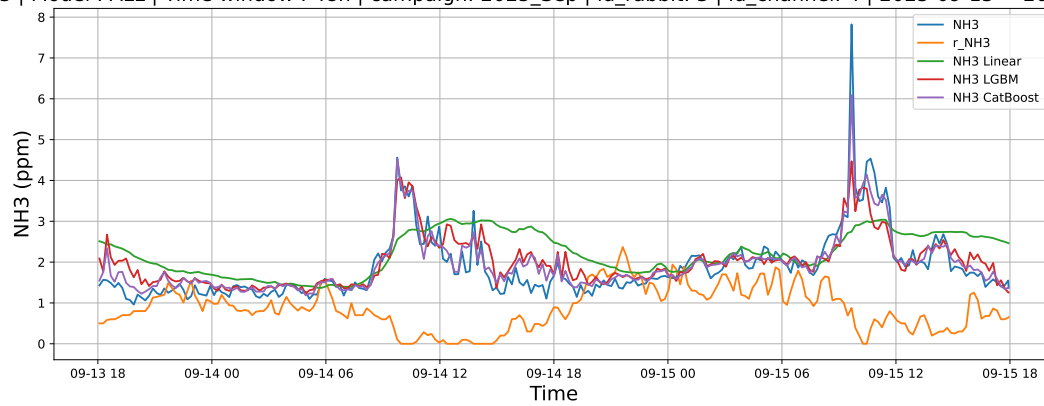
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



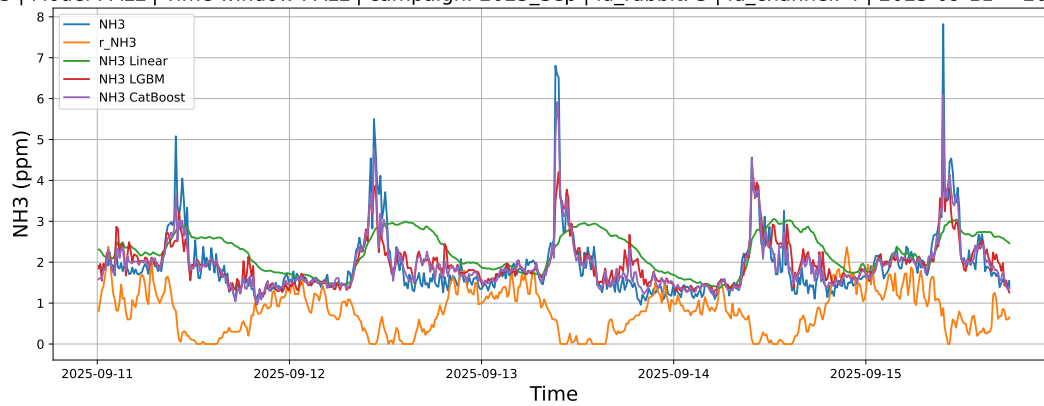
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

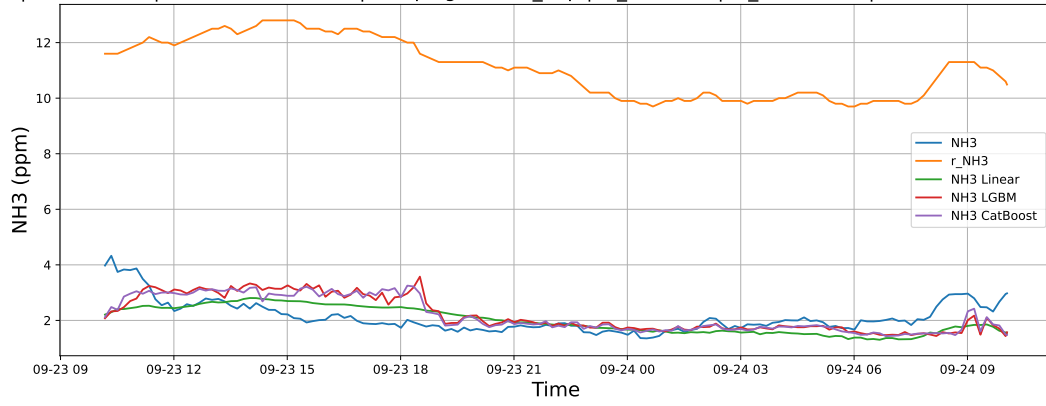
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

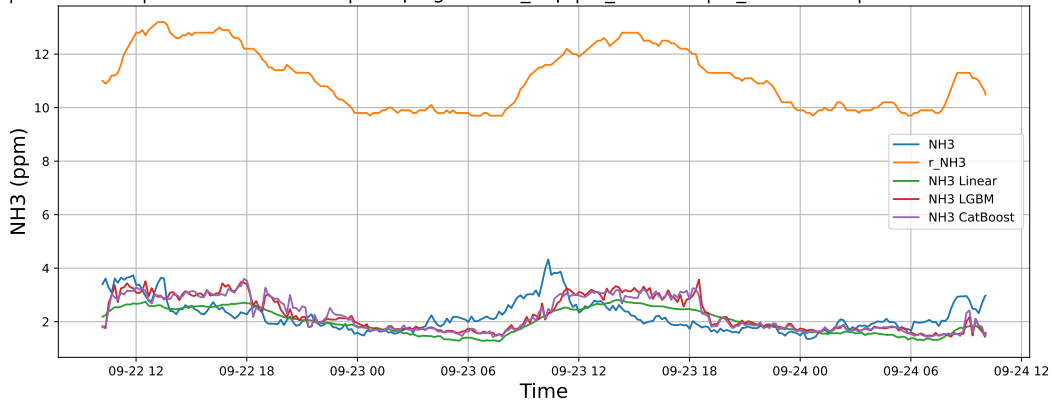
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



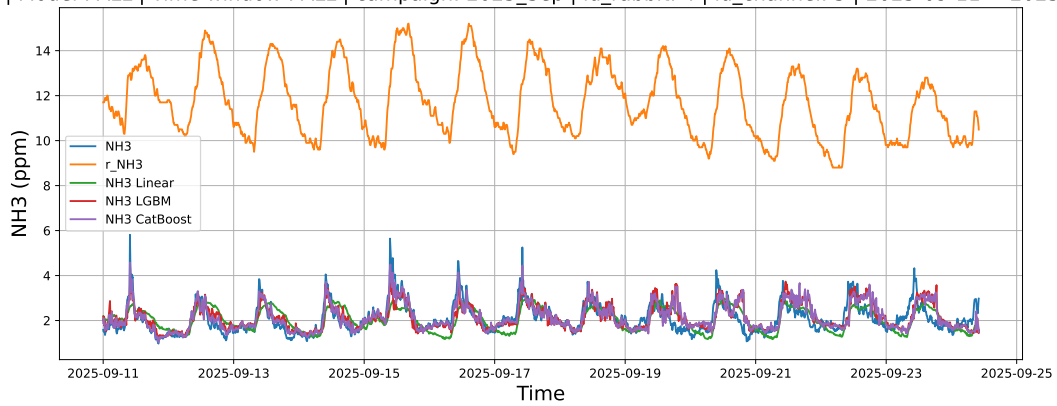
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

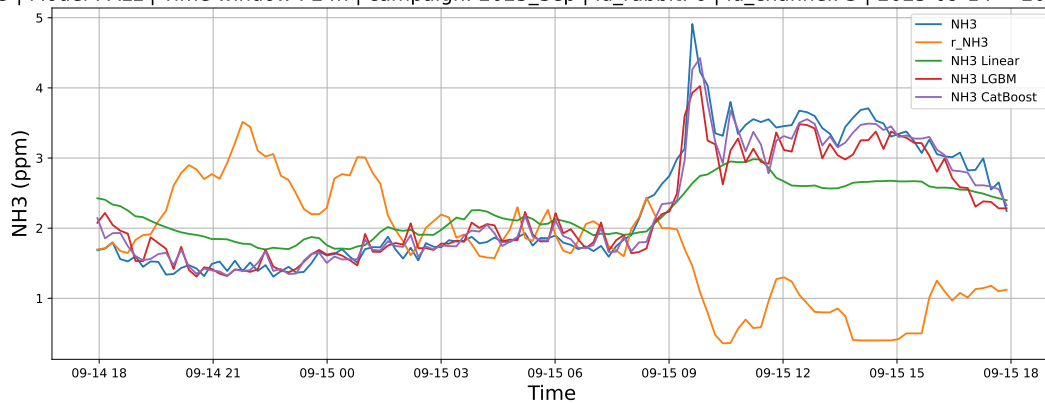
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

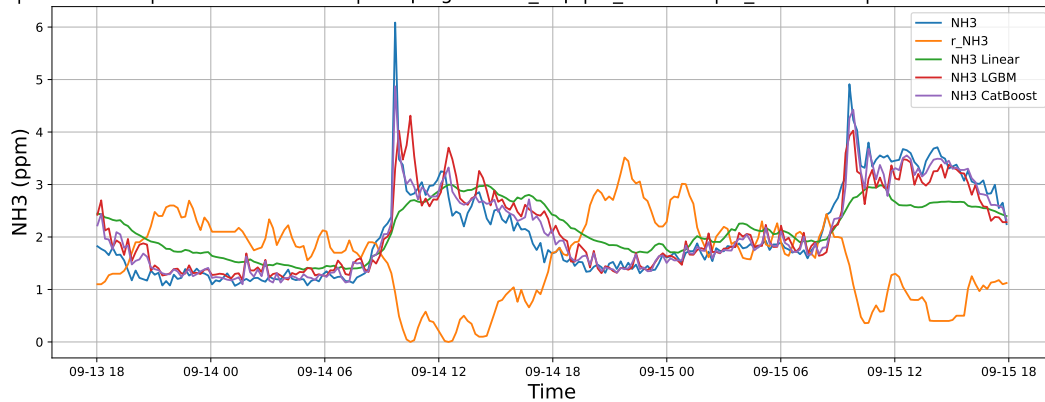
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

